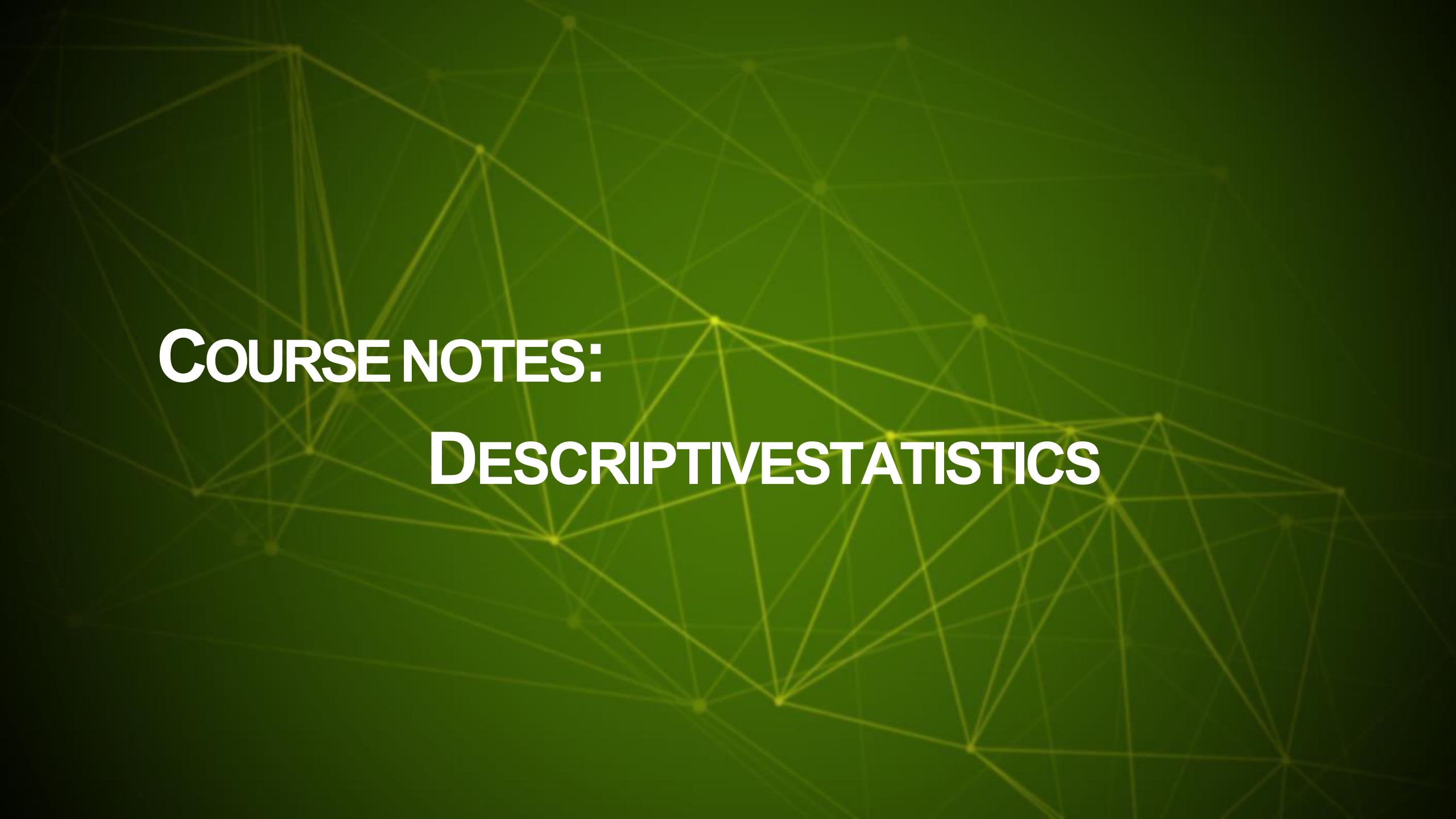




COURSE NOTES:

DESCRIPTIVE STATISTICS

FIRST WE NEED TO KNOW THE VARIABLES ...



TYPES OF DATA

```
graph TD; A[TYPES OF DATA] --> B[CATEGORICAL]; A --> C[NUMERICAL]; D[MEASUREMENT LEVELS];
```

***MEASUREMENT
LEVELS***

CATEGORICAL

NUMERICAL



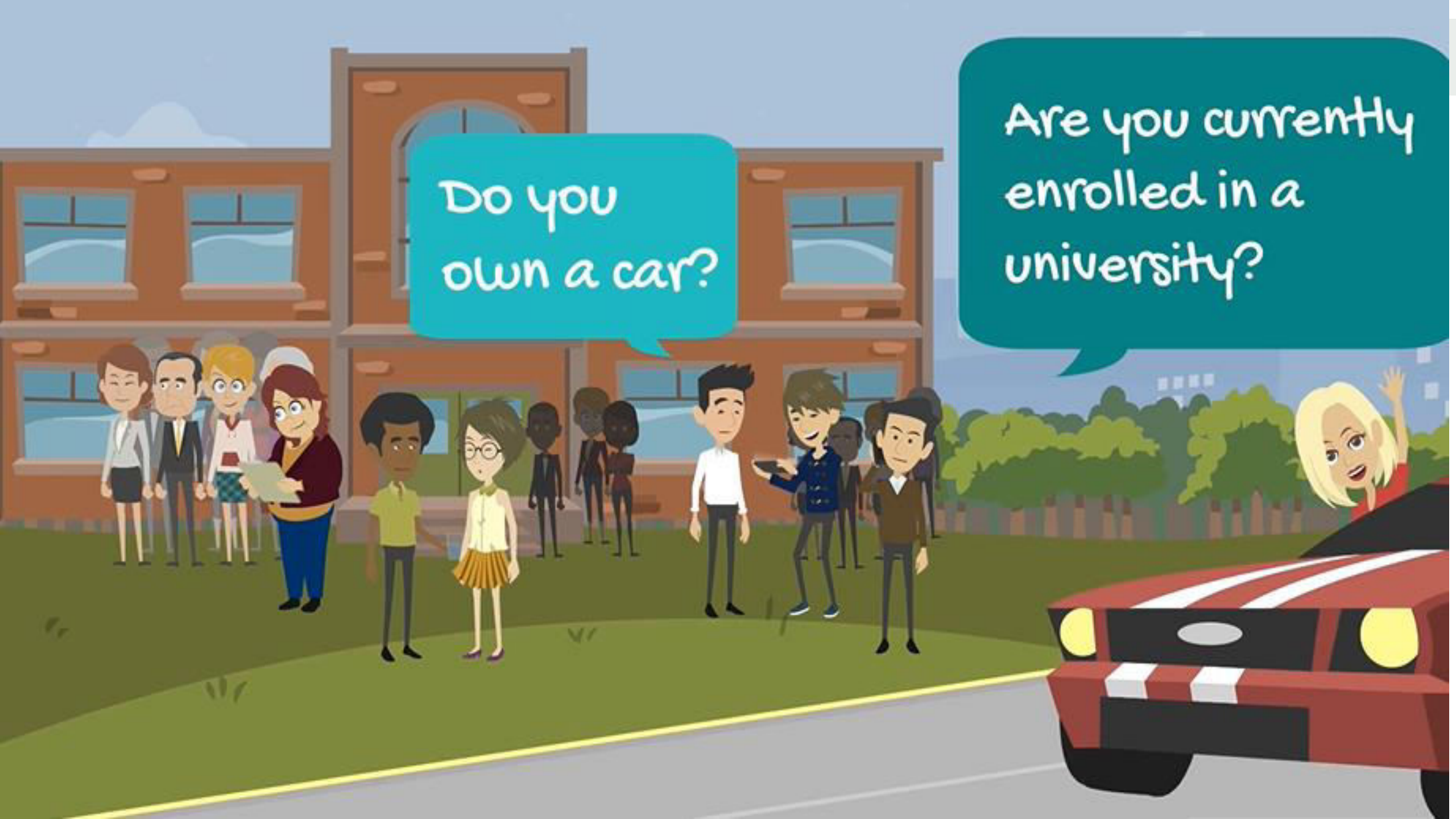
CATEGORICAL

categories or groups

CAR BRANDS



YES AND NO QUESTIONS



Do you
own a car?

Are you currently
enrolled in a
university?

TYPES OF DATA

```
graph TD; A[TYPES OF DATA] --> B[CATEGORICAL]; A --> C[NUMERICAL]; D[MEASUREMENT LEVELS];
```

***MEASUREMENT
LEVELS***

CATEGORICAL

NUMERICAL

TYPES OF DATA

MEASUREMENT LEVELS

CATEGORICAL

NUMERICAL

DISCRETE

CONTINUOUS



DISCRETE

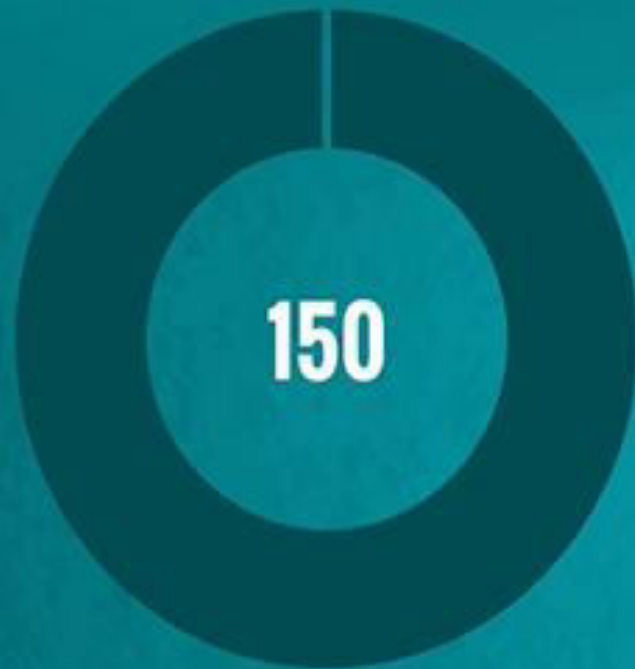
of children

S.A.T score



1000; 1560; 1570; 2400

CONTINUOUS



68.0389 kg



CONTINUOUS

Weight



DISCRETE

of children



EXAMPLES OF DISCRETE

A, B, C,
D, E, F
or
0 to 100%

Grades



#1,000

number of objects



Money

EXAMPLES OF CONTINUOUS



Height



Area



Distance



Time



***TIME ON A CLOCK IS
DISCRETE***



***TIME IN GENERAL IS
CONTINUOUS***

A variable represents the weight of a person. What type of data does it represent?

- 1. Categorical, discrete**
- 2. Categorical, continuous**
- 3. Numerical, discrete**
- 4. Numerical, continuous**

Answer is

numerical, continous

**A variable represents the gender of a person.
What type of data does it represent ?**

- 1. Categorical**
- 2. numerical, discrete**
- 3. Numerical, continuous**

Answer is

Categorical

MEASUREMENT LEVELS

```
graph TD; A[MEASUREMENT LEVELS] --> B[QUALITATIVE]; A --> C[QUANTITATIVE]; B --> D[NOMINAL]; B --> E[ORDINAL]; C --> F[INTERVAL]; C --> G[RATIO];
```

QUALITATIVE

QUANTITATIVE

NOMINAL

ORDINAL

INTERVAL

RATIO

**MEASUREMENT
LEVELS**

```
graph TD; A[MEASUREMENT LEVELS] --> B[QUALITATIVE]; A --> C[QUANTITATIVE]; B --> D[NOMINAL]; B --> E[ORDINAL];
```

QUALITATIVE

QUANTITATIVE

NOMINAL

ORDINAL

NOMINAL



1. Which of the following courses would you like to register for? (Choose any 3)

- Web development
- Product design
- Graphics design
- Content writing
- Animation

II. What toppings would you like on your pizza?

- Pepperoni
- Sausage
- Spinach
- Sardines
- Extra Cheese

ORDINAL



Disgusting

Unappetizing

Neutral

Tasty

Delicious



Ordinal Data Collection Tool

Which of the following best describes your current level of financial happiness?

- ☐ Very happy
 - ☐ Happy
 - ☒ Neutral
 - ☐ Unhappy
 - ☐ Very unhappy
-

Ordinal Data Collection Tool

How can you rate your knowledge of Excel?

- ☐ Advanced
- ☒ Intermediate
- ☐ Basic
- ☐ Novice
- ☐ Zero

MEASUREMENT LEVELS

```
graph TD; A[MEASUREMENT LEVELS] --> B[QUALITATIVE]; A --> C[QUANTITATIVE]; B --> D[NOMINAL]; B --> E[ORDINAL]; C --> F[INTERVAL]; C --> G[RATIO];
```

QUALITATIVE

QUANTITATIVE

NOMINAL

ORDINAL

INTERVAL

RATIO

RATIO

HAS A TRUE



INTERVAL

DOESN'T HAVE A TRUE



RATIO



x2



x6

You have 3 times
as many as I do.

RATIO OF 6/2 IS

3

RATIO



***NUMBER OF
OBJECTS***



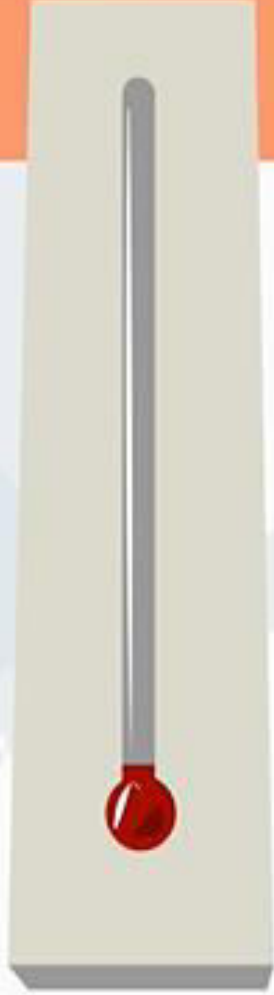
DISTANCE



TIME

INTERVAL

DOESN'T HAVE A TRUE



INTERVAL

TODAY

seems colder in
 $^{\circ}\text{C}$

in terms of
 $^{\circ}\text{F}$... not really



INTERVAL

0[°]C AND 0[°]F ARE NOT TRUE ZEROS

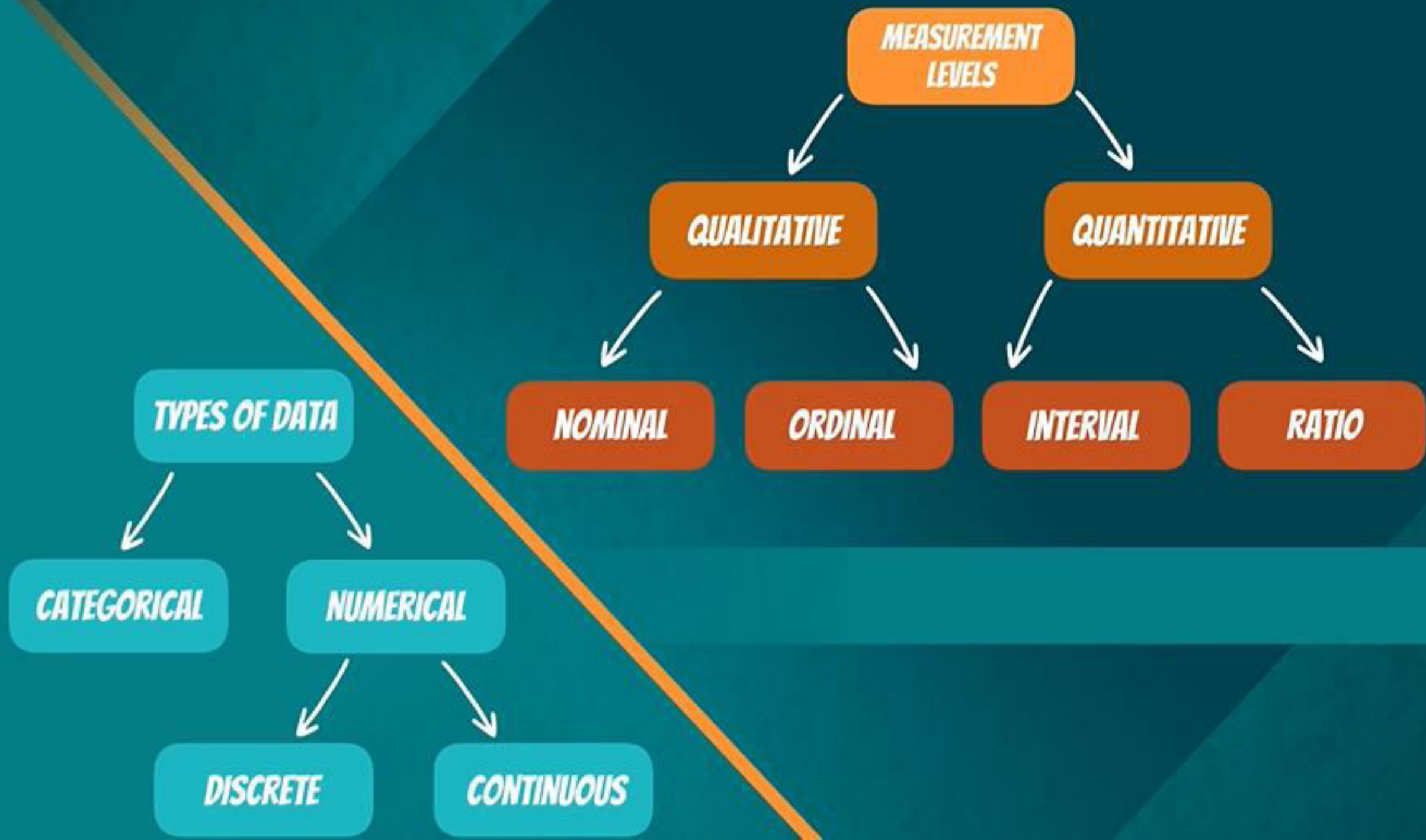


INTERVAL

0°C AND 0°F ARE NOT TRUE ZEROS

$$0^{\circ}\text{K} = -273.15^{\circ}\text{C} = -459.67^{\circ}\text{F}$$





Question 1:

A variable represents the gender of a person. What is the level of measurement?

☐ nominal

☐ ordinal

☐ interval

☐ ratio

Question 2:

A variable represents the weight of a person. What is the level of measurement?

☐ nominal

☐ ordinal

☐ interval

☐ ratio

***VISUALIZING DATA IS THE MOST INTUITIVE
WAY TO INTERPRET IT***





TYPES OF DATA

***MEASUREMENT
LEVELS***

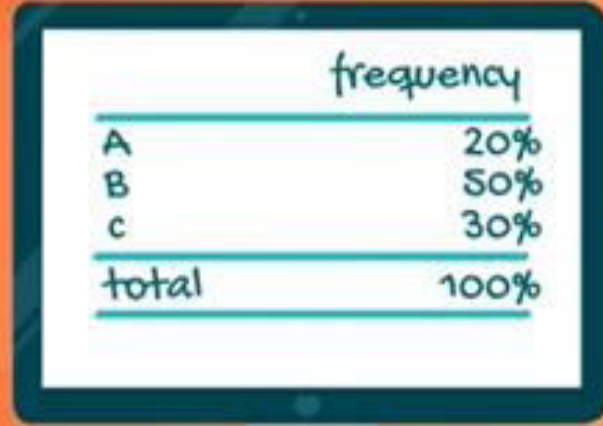
CATEGORICAL

NUMERICAL



REPRESENTATION OF CATEGORICAL VARIABLES

Frequency
distribution
tables

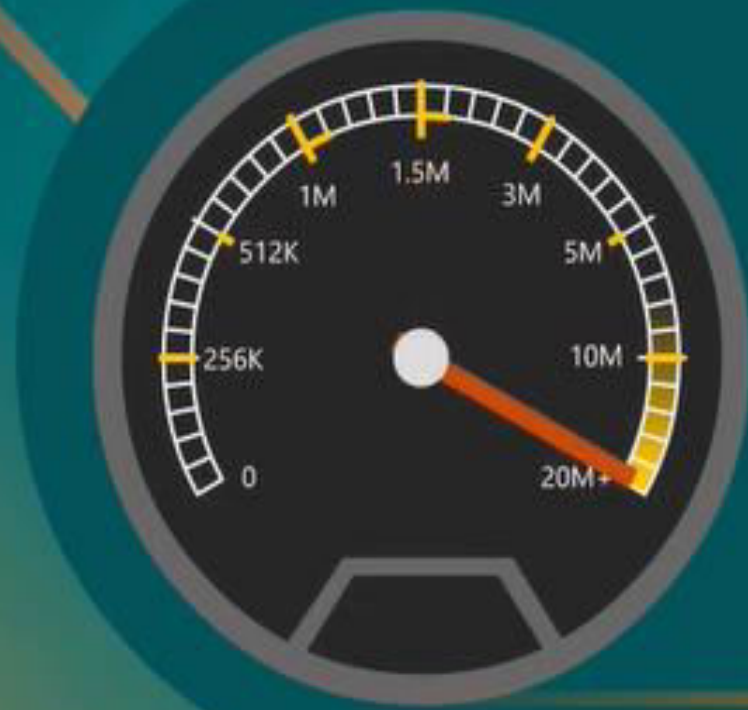


Bar charts

Pie charts

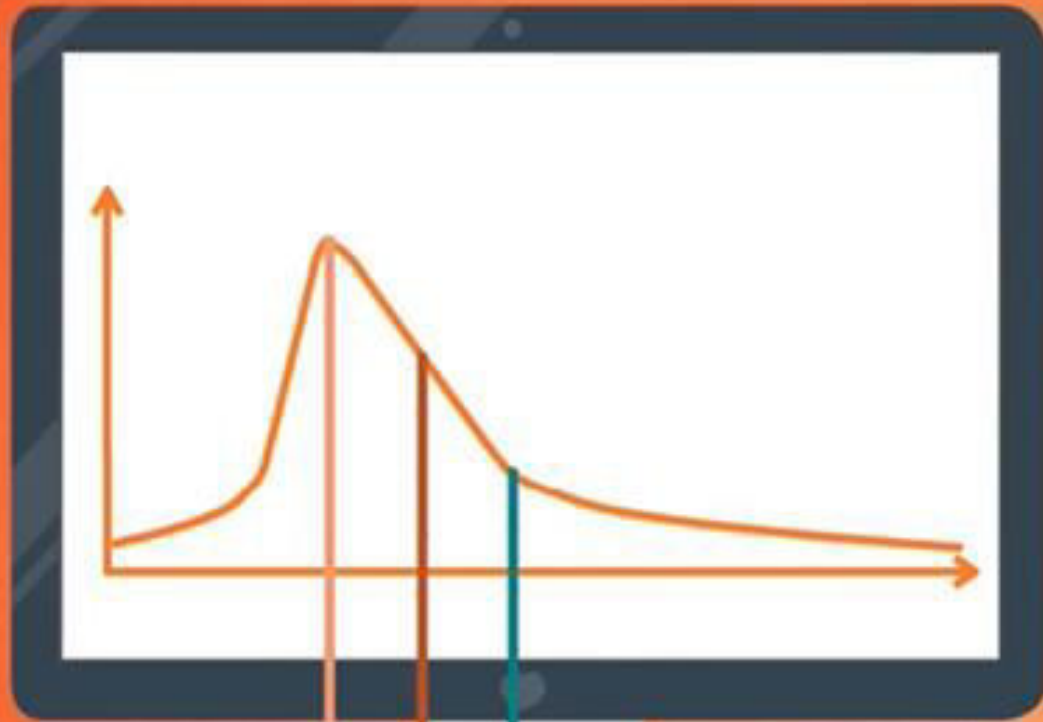


Pareto
diagrams



MEASURES OF CENTRAL TENDENCY

MEASURES OF CENTRAL TENDENCY



Mean

Median

Mode

MEASURES OF CENTRAL TENDENCY

D.

Test

a.k.a. simple
average

MEAN

Population μ

Sample $\bar{x}$

HOW DO WE FIND THE MEAN

$$\frac{\sum_{i=1}^N x_i}{N}$$

By adding up all the components and then dividing by the number of components

or

$$\frac{x_1 + x_2 + x_3 + \dots + x_{N-1} + x_N}{N}$$

1	Mean, median, mode
---	--------------------

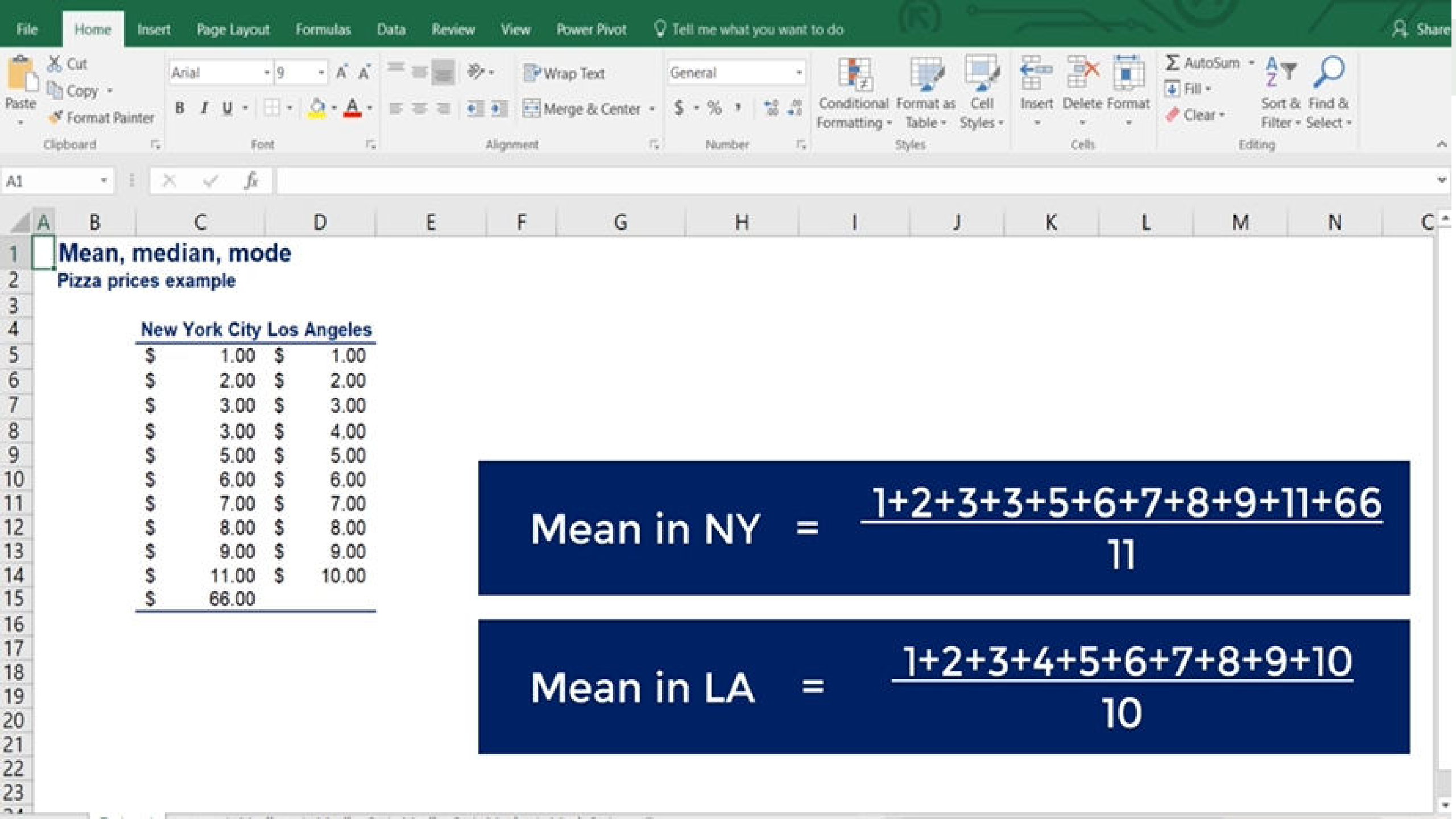
3
4 New York City Los Angeles

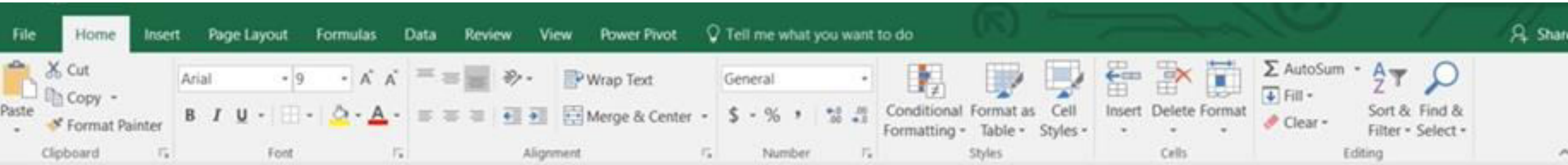
6	\$	2.00	\$	2.00
---	----	------	----	------

8	\$	3.00	\$	4.00
9	\$	5.00	\$	5.00

11	\$	7.00	\$	7.00
12	\$	8.00	\$	8.00

14	\$	11.00	\$	10.00
----	----	-------	----	-------





Mean, median, mode

Pizza prices example

Position	New York City	Los Angeles
1	\$ 1.00	\$ 1.00
2	\$ 2.00	\$ 2.00
3	\$ 3.00	\$ 3.00
4	\$ 3.00	\$ 4.00
5	\$ 5.00	\$ 5.00
6	\$ 6.00	\$ 6.00
7	\$ 7.00	\$ 7.00
8	\$ 8.00	\$ 8.00
9	\$ 9.00	\$ 9.00
10	\$ 11.00	\$ 10.00
11	\$ 66.00	

	New York City	Los Angeles
Mean	\$ 11.00	\$ 5.50
Median		

The median is the number at position $(n+1)/2$ in the ordered list

1 Mean, median, mode

2 Pizza prices example

Position	New York City	Los Angeles
1	\$ 1.00	\$ 1.00
2	\$ 2.00	\$ 2.00
3	\$ 3.00	\$ 3.00
4	\$ 3.00	\$ 4.00
5	\$ 5.00	\$ 5.00
6	\$ 6.00	\$ 6.00
7	\$ 7.00	\$ 7.00
8	\$ 8.00	\$ 8.00
9	\$ 9.00	\$ 9.00
10	\$ 11.00	\$ 10.00
11	\$ 66.00	

	New York City	Los Angeles
Mean	\$ 11.00	\$ 5.50
Median	\$ 6.00	

Median in NYC = $(11+1)/2 = 6$ th position

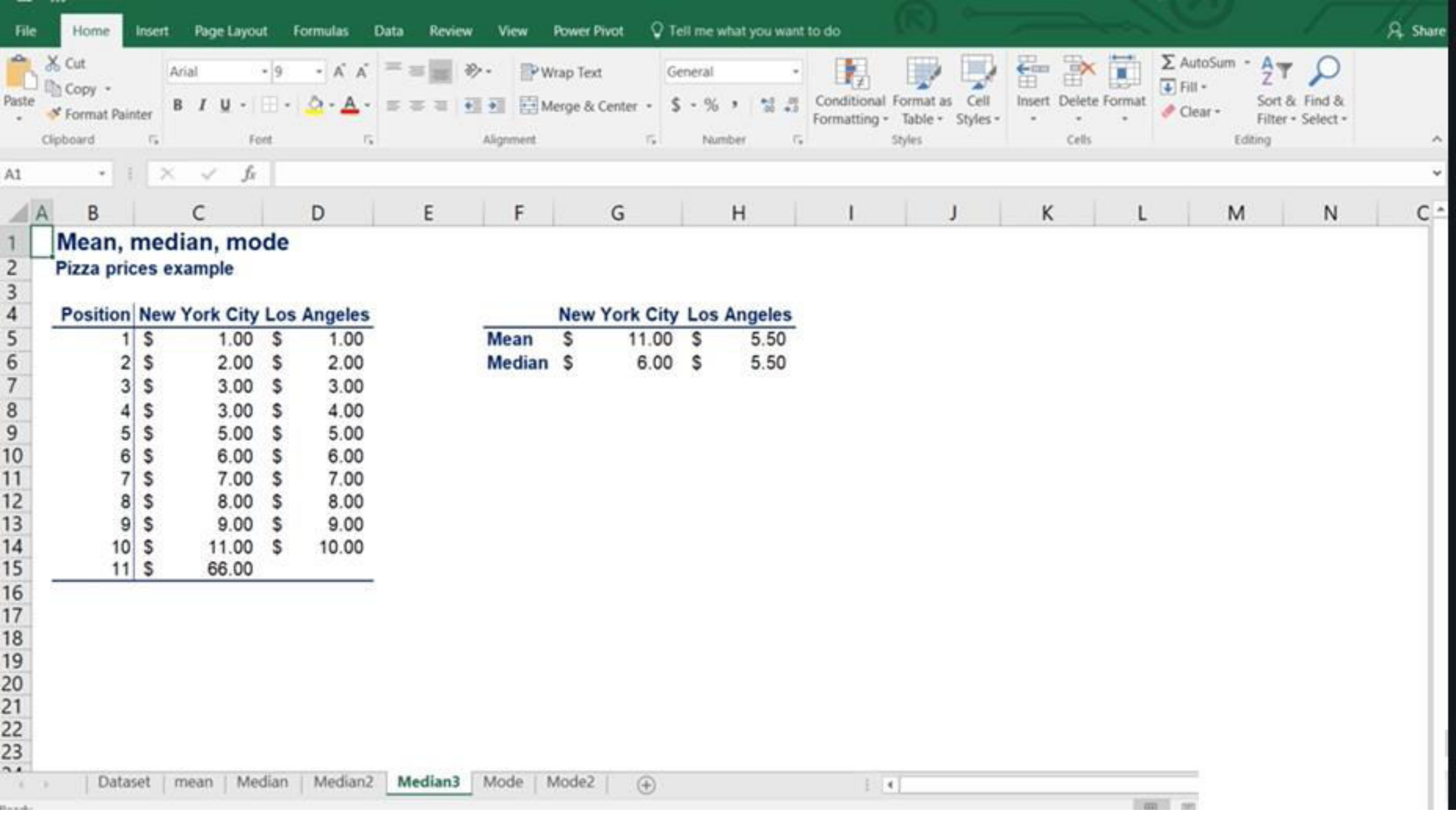


1	Mean, median, mode
2	Pizza prices example
3	
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22	
23	

Position	New York City	Los Angeles
1	\$ 1.00	\$ 1.00
2	\$ 2.00	\$ 2.00
3	\$ 3.00	\$ 3.00
4	\$ 3.00	\$ 4.00
5	\$ 5.00	\$ 5.00
6	\$ 6.00	\$ 6.00
7	\$ 7.00	\$ 7.00
8	\$ 8.00	\$ 8.00
9	\$ 9.00	\$ 9.00
10	\$ 11.00	\$ 10.00
11	\$ 66.00	

	New York City	Los Angeles
Mean	\$ 11.00	\$ 5.50
Median	\$ 6.00	

Median in LA = $(10+1)/2 = 5.5$ th position





Mean, median, mode

Pizza prices example

Position	New York City	Los Angeles
1	\$ 1.00	\$ 1.00
2	\$ 2.00	\$ 2.00
3	\$ 3.00	\$ 3.00
4	\$ 3.00	\$ 4.00
5	\$ 5.00	\$ 5.00
6	\$ 6.00	\$ 6.00
7	\$ 7.00	\$ 7.00
8	\$ 8.00	\$ 8.00
9	\$ 9.00	\$ 9.00
10	\$ 11.00	\$ 10.00
11	\$ 66.00	

	New York City	Los Angeles
Mean	\$ 11.00	\$ 5.50
Median	\$ 6.00	\$ 5.50

The mode is the value that occurs most often

File

Home

Insert

Page Layout

Formulas

Data

Review

View

Power Pivot

Tell me what you want to do

Share

Cut

Copy

Format Painter

Arial

- 9

A⁺

A⁻

Wrap Text

Merge & Center

\$

- %

,

1/2

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100

General

Conditional Formatting

Format as Table

Cell Styles

Insert

Delete

Format

AutoSum

Fill

Clear

Sort & Filter

Find & Select

Clipboard

Font

Alignment

Number

Styles

Cells

Editing

A1

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M

N

C

1

Mean, median, mode

2

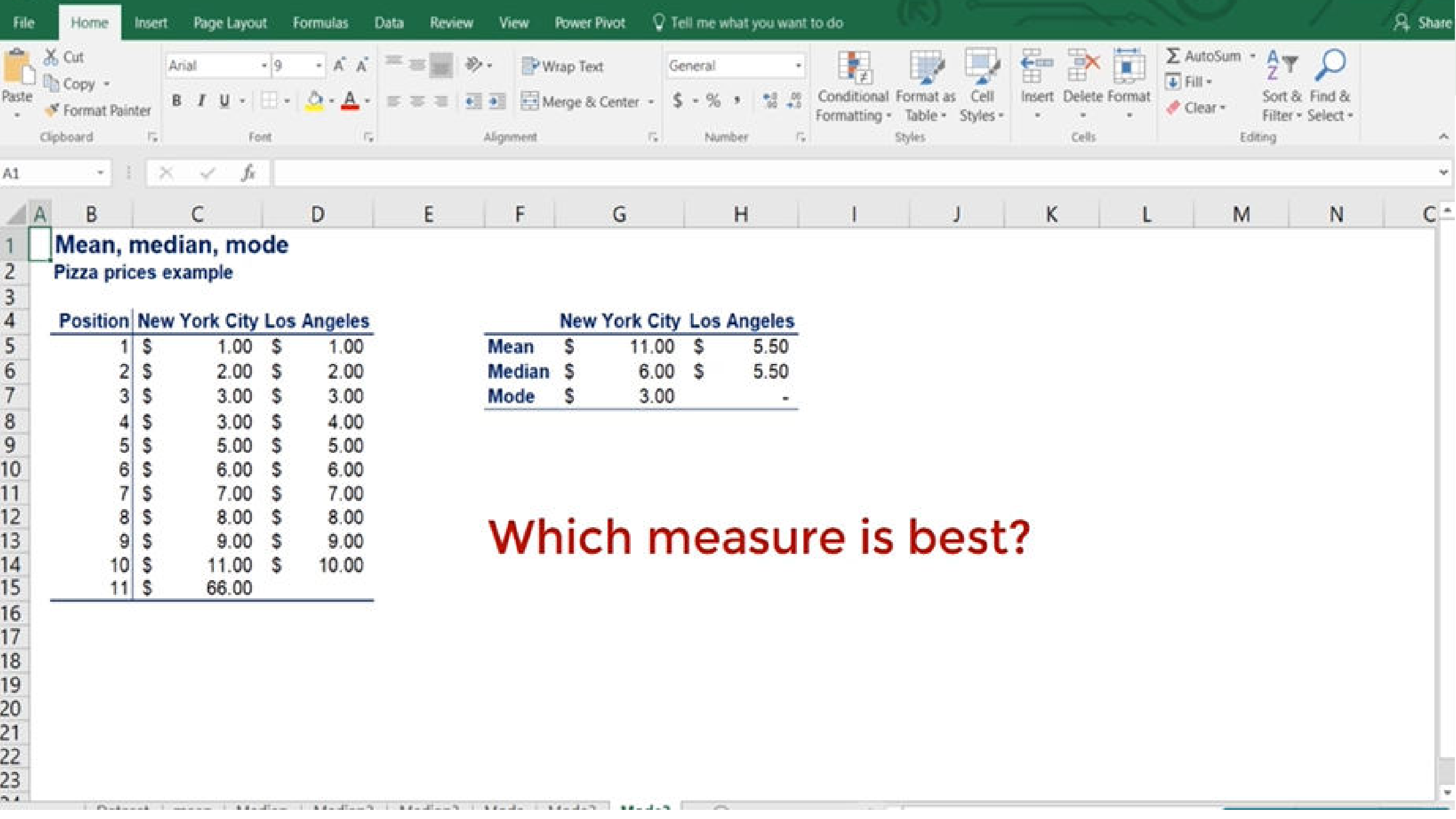
Pizza prices example

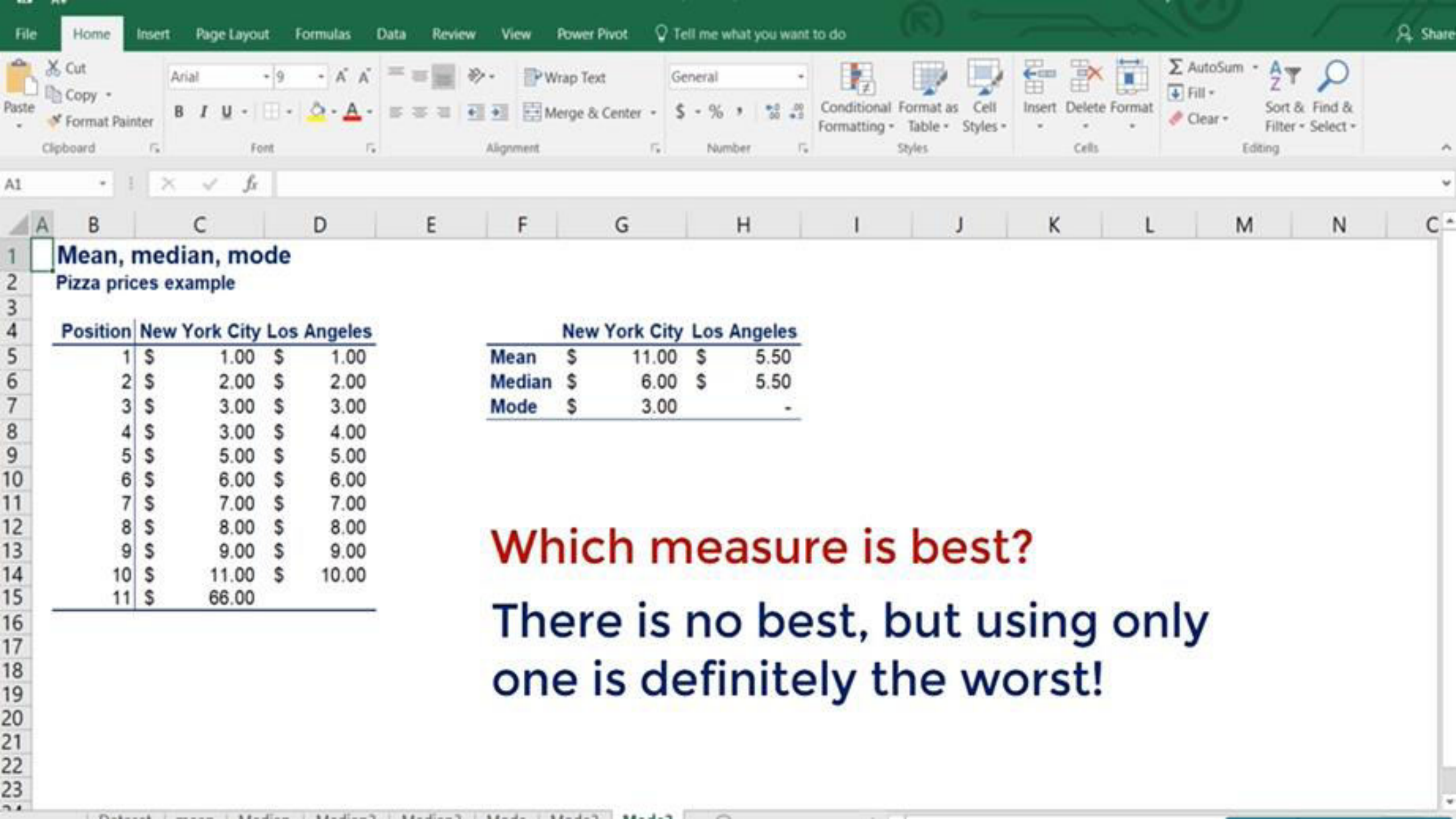
3

4

Position	New York City	Los Angeles
1	\$ 1.00	\$ 1.00
2	\$ 2.00	\$ 2.00
3	\$ 3.00	\$ 3.00
4	\$ 3.00	\$ 4.00
5	\$ 5.00	\$ 5.00
6	\$ 6.00	\$ 6.00
7	\$ 7.00	\$ 7.00
8	\$ 8.00	\$ 8.00
9	\$ 9.00	\$ 9.00
10	\$ 11.00	\$ 10.00
11	\$ 66.00	

	New York City	Los Angeles
Mean	\$ 11.00	\$ 5.50
Median	\$ 6.00	\$ 5.50
Mode	\$ 3.00	



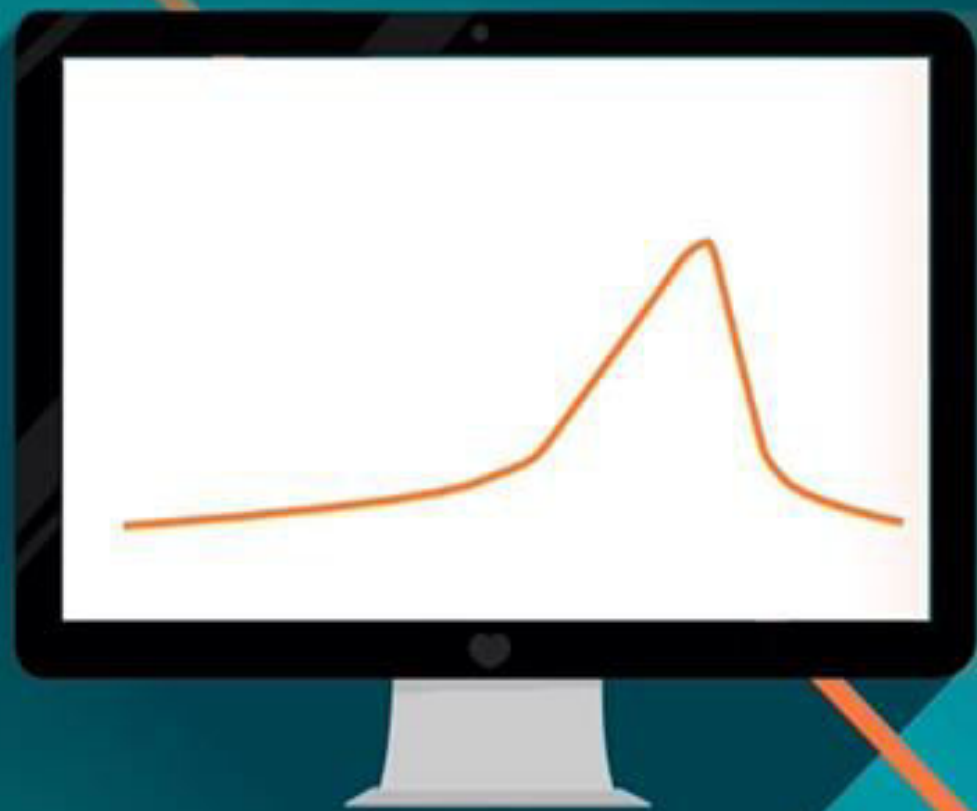


***MEASURES OF
CENTRAL TENDENCY***

***MEASURES OF
ASYMMETRY***

SAMPLE SKEWNESS FORMULA

$$\frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^3}{\sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}}^3$$



Almost always, you will
use software that
performs the calculation
for you



Skewness indicates
whether the data is
concentrated on one side

Skewness

Positive (right)

Dataset 1	Interval	Frequency	Dataset 2	Interval	Frequency	Dataset 3	Interval	Frequency
1	0 to 1	4	1	0 to 1	2	1	0 to 1	1
1	1 to 2	6	1	1 to 2	2	2	1 to 2	1
1	2 to 3	4	2	2 to 3	3	3	2 to 3	2
1	3 to 4	2	2	3 to 4	3	3	3 to 4	3
2	4 to 5	2	3	4 to 5	3	4	4 to 5	4
2	5 to 6	0	3	5 to 6	2	4	5 to 6	6
2	6 to 7	1	3	6 to 7	2	4	6 to 7	2
2			4			5		
2			4			5		
2			4			5		
3			4			5		
3			4			6		
3			4			6		
3			4			6		
4			5			6		
4			5			6		
5			6			6		
5			6			6		
6			7			7		
7			7			7		

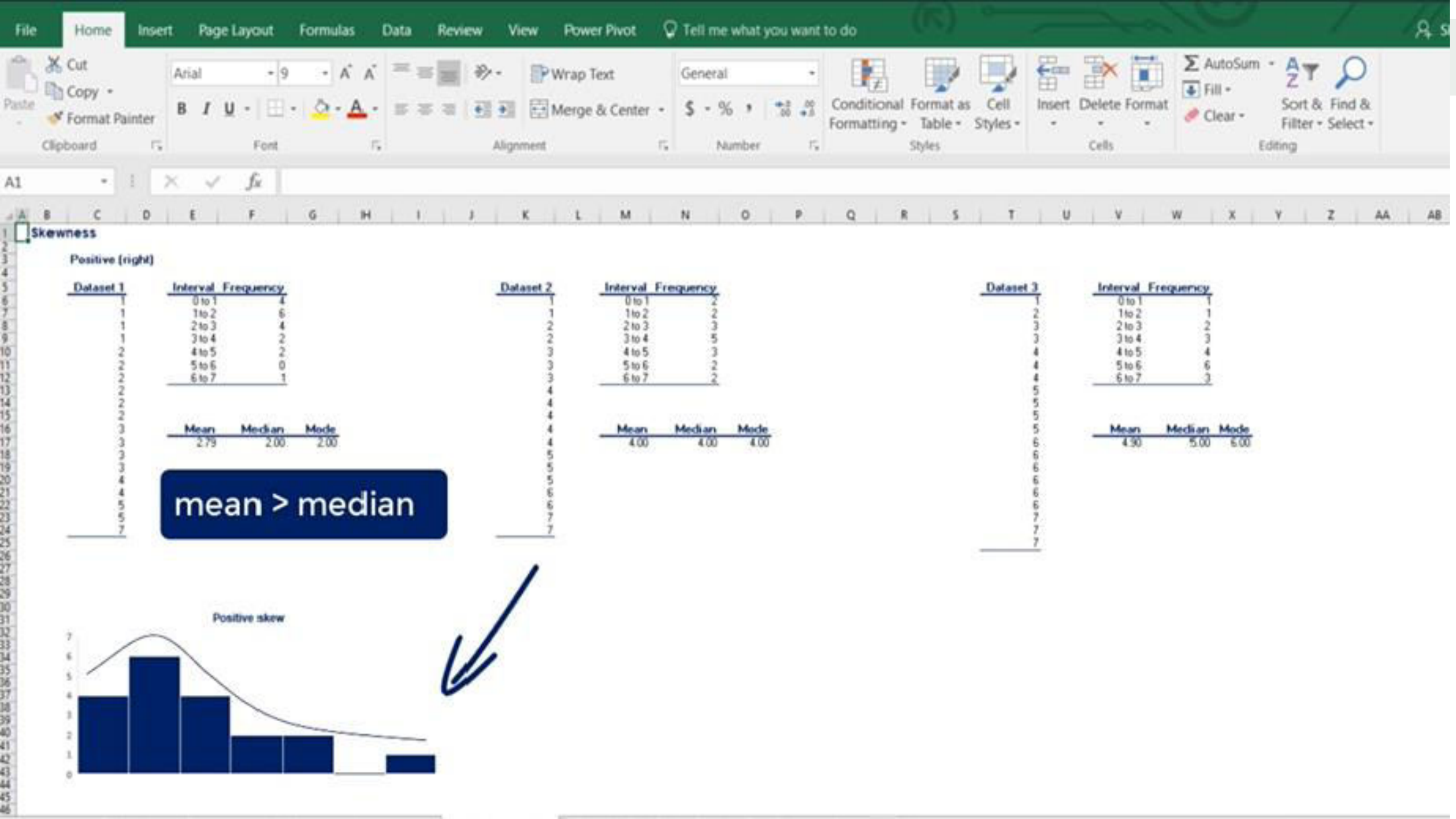
Mean	Median	Mode
2.79	2.00	2.00

Mean	Median	Mode
4.00	4.00	4.00

Mean	Median	Mode
4.90	5.00	6.00

mean > median





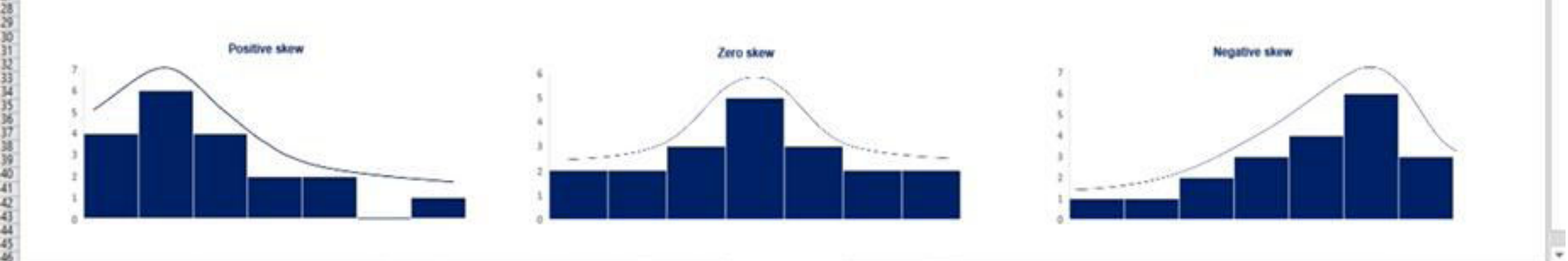
A1 X Y Z AA AB

Skewness									
Positive (right)					Zero (no skew)				
Dataset 1		Interval Frequency			Dataset 2		Interval Frequency		
1	1	0 to 1	4		1	1	0 to 1	2	
2	1	1 to 2	6		2	1	1 to 2	2	
3	1	2 to 3	4		3	2	2 to 3	3	
4	1	3 to 4	2		4	2	3 to 4	5	
5	2	4 to 5	2		5	3	4 to 5	3	
6	2	5 to 6	0		6	3	5 to 6	2	
7	2	6 to 7	1		7	3	6 to 7	2	
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205	3				205	4			
206	3				206	4			

Skewness

2025 RELEASE UNDER E.O. 14176

mean < median



**Measuring how data is spread out :
calculating variance**

MEASURES OF

Central tendency

Asymmetry

Variability



WE WILL COVER

> **VARIANCE**

> **STANDARD
DEVIATION**

> **COEFFICIENT OF
VARIATION**

MEASURES OF

Central tendency

Asymmetry

Variability

DIFFERENT FORMULAS FOR

SAMPLE DATA



POPULATION DATA



WHEN YOU TAKE SAMPLE DATA

SAMPLE DATA



A sample statistic is an approximation of the population parameter

WHEN YOU TAKE SAMPLE DATA



SAMPLE DATA

10 different samples give

10 different measures

WHEN YOU TAKE SAMPLE DATA

FORMULA



```
graph LR; A[FORMULA] --> B[population formula]; A --> C[sample formula];
```

The diagram illustrates the relationship between a general formula and its specific applications. A central box labeled 'FORMULA' has two curved arrows pointing to the right. The top arrow points to a box labeled 'population formula', and the bottom arrow points to a box labeled 'sample formula'. All boxes are teal with white text, set against an orange background.

population formula

sample formula

Mean

$$\frac{\sum_{i=1}^n x_i}{n}$$

sample formula

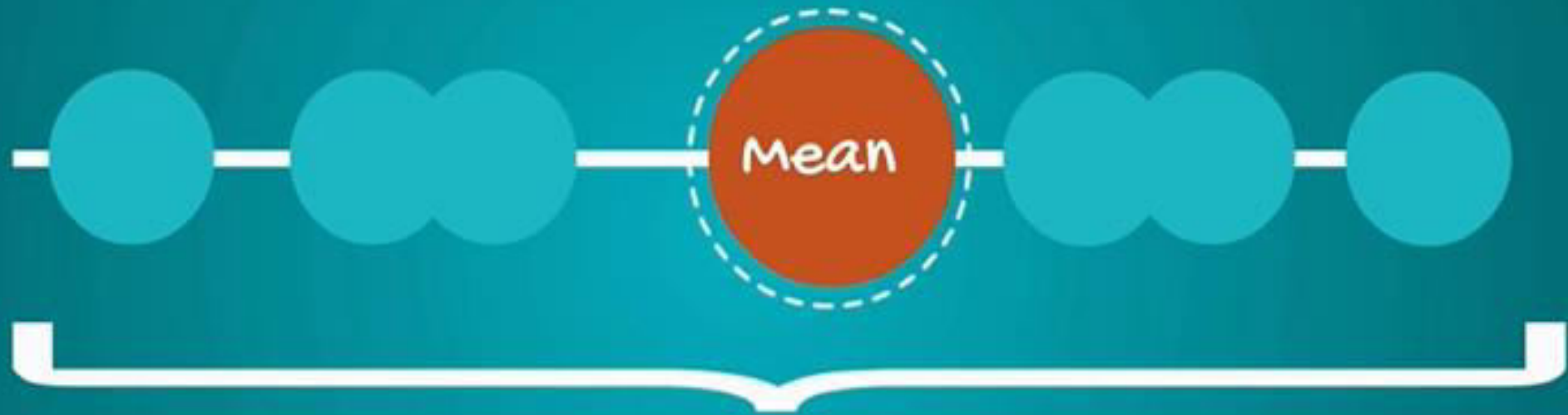
n is the size of
the sample

$$\frac{\sum_{i=1}^N x_i}{N}$$

population formula

N is the size of
the population

VARIANCE



Variance measures the dispersion of a set of data points around their mean

VARIANCE

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$



population
variance

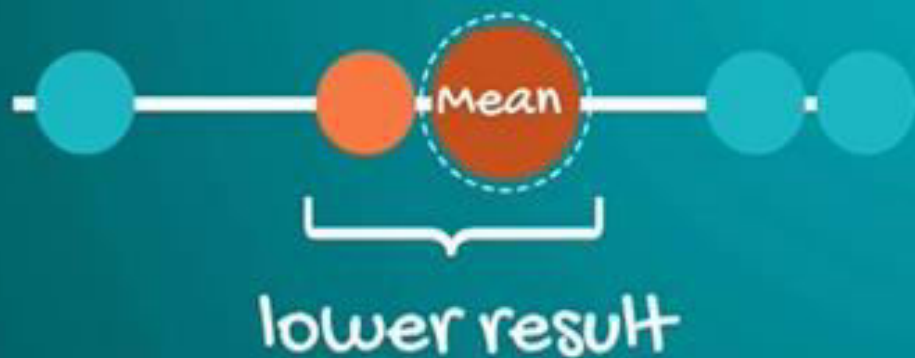
$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$



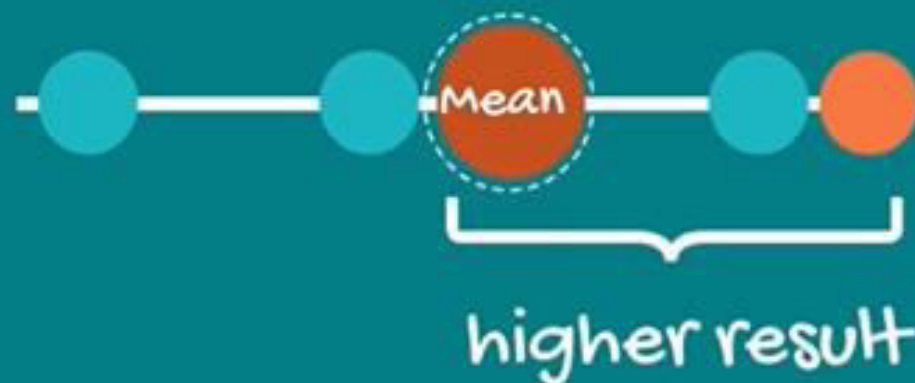
sample
variance

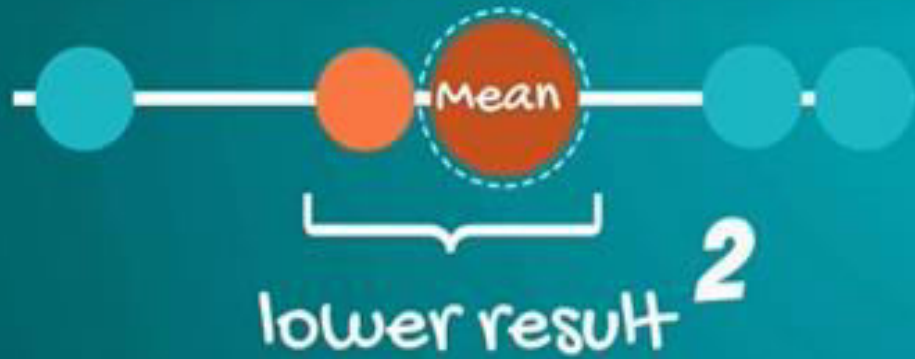
VARIANCE

$$\sigma^2 = \frac{\sum_{i=1}^N (x_i - \mu)^2}{N}$$



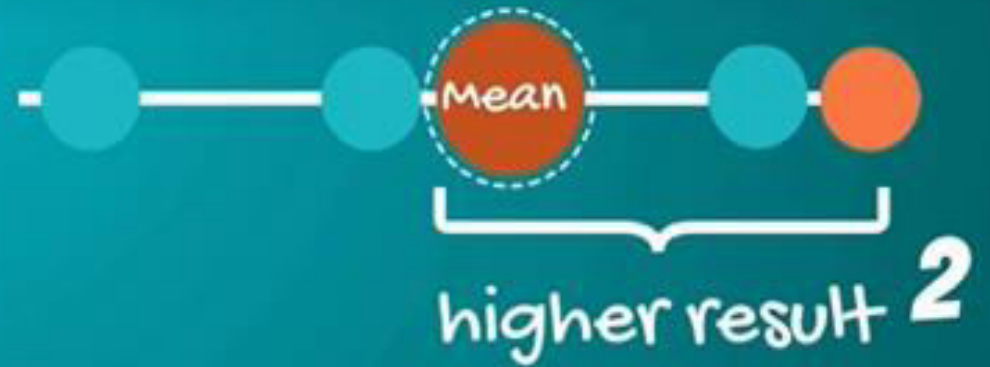
population
variance





➤ Dispersion is non-negative.
Non-negative values don't cancel out

➤ Amplifies the effect of large differences

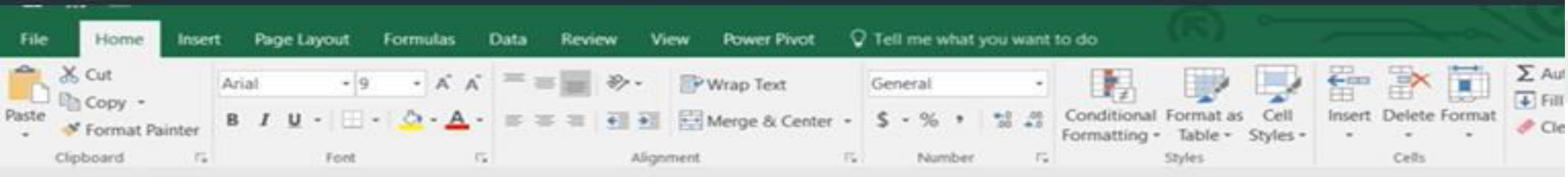


	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1		Variance															
2																	
3		Population															
4		1	Mean		3.00												
5		2															
6		3															
7		4															
8		5															
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29																	

We start by calculating the mean. $\text{Mean} = \frac{1+2+3+4+5}{5}$

Population size

Population variance formula



A1															
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1		Variance													
2															
3		Population													
4		1	Mean		3.00										
5		2	Population variance		2.00										
6		3													
7		4													
8		5													
9															
10															
11															
12															
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30															
31															

Sample variance formula is used when our set of observations is a sample drawn from a bigger population



A1 X ✓ fx

A B C D E F G H I J K L M N O P Q R

1	Variance																		
2																			
3	Population																		
4	1	Mean		3.00															
5	2	Population variance		2.00															
6	3																		
7	4																		
8	5																		
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30																			
31																			
32																			

$$\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = \frac{(1-3)^2 + (2-3)^2 + (3-3)^2 + (4-3)^2 + (5-3)^2}{4}$$

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Conditional Formatting

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Find & Select

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31

Variance

Population

1

2

3

4

5

Mean

Population variance

Sample variance

3.00

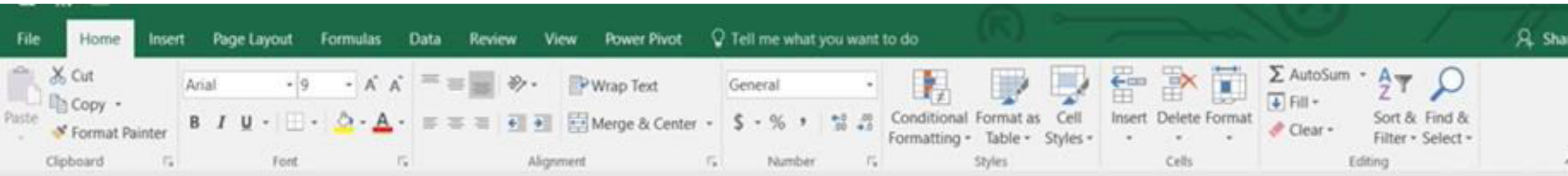
2.00

2.50

We had all the data and we calculated the variance.

We had a sample, but did not know the population.

Therefore, there is more uncertainty.



Variance																			
Population																			
Mean																			
Population variance																			
Sample variance																			
1	3.00	2.00	2.50	1	3.20	2.96													
2				1															
3				1															
4				2															
5				3															
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30				5															
31				5															

Our sample variance has rightfully corrected upwards, in order to reflect the higher potential variability



WE WILL COVER

> **VARIANCE**

> **STANDARD
DEVIATION**

> **COEFFICIENT OF
VARIATION**

VARIANCE

large
units 2





WE WILL COVER

> **VARIANCE**

> **STANDARD
DEVIATION**

> **COEFFICIENT OF
VARIATION**

VARIANCE

large
units 2



WHEN YOU TAKE SAMPLE DATA

FORMULA



population formula



sample formula

STANDARD DEVIATION FORMULAS

population standard
deviation

sample standard
deviation

STANDARD DEVIATION FORMULAS

$$\sigma = \sqrt{\sigma^2}$$

population standard
deviation

sample standard
deviation

$$S = \sqrt{S^2}$$

COEFFICIENT OF VARIATION (CV)

/relative standard
deviation/

standard deviation

mean

COEFFICIENT OF VARIATION (CV)

$$c_v = \frac{\sigma}{\mu}$$

Population formula

sample formula

$$\hat{c}_v = \frac{s}{\bar{x}}$$



σ

Standard deviation is the most common measure of variability for a SINGLE DATASET



c_v

A white Greek letter sigma (σ) is centered within a solid orange circle. This circle is surrounded by a larger teal circle with a dashed white border.

σ

Standard deviation is the most common measure of variability for a SINGLE DATASET

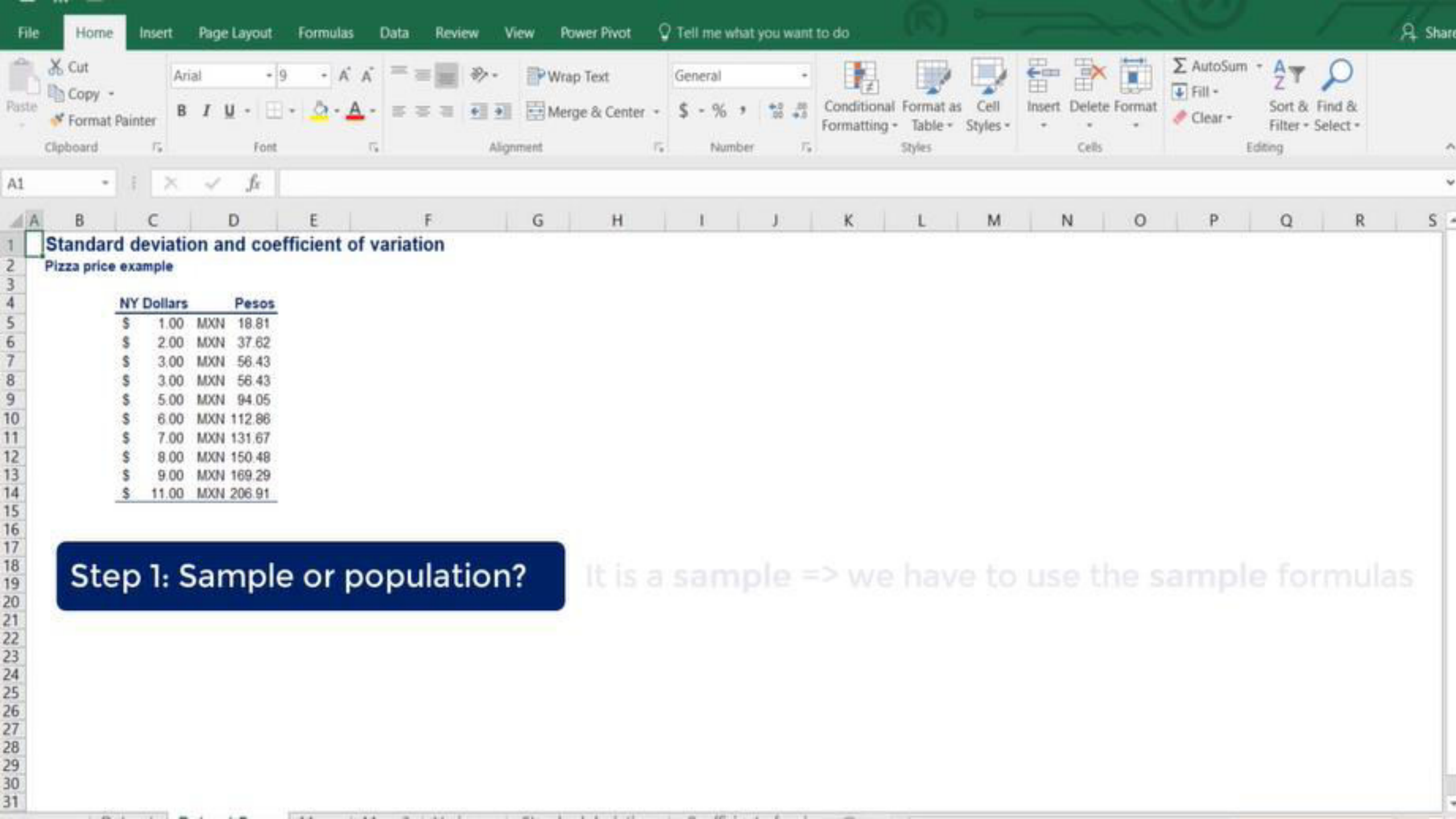
Comparing TWO OR MORE datasets

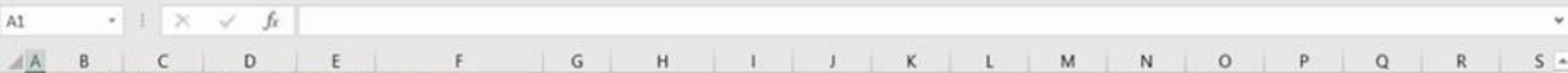
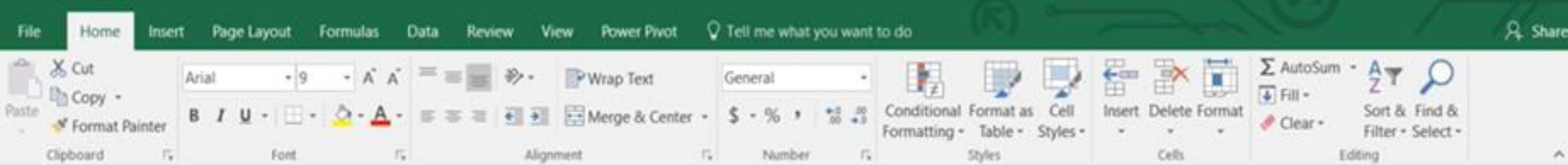
A white symbol C_v is centered within a solid orange circle. This circle is surrounded by a larger teal circle with a dashed white border.

C_v

Standard deviation and coefficient of variation

18





1 **Standard deviation and coefficient of variation**
2 **Pizza price example**

NY Dollars		Pesos				Dollars		Pesos	
\$	1.00	MXN	18.81	Mean		\$	5.50	MXN	103.46
\$	2.00	MXN	37.62						
\$	3.00	MXN	56.43						
\$	3.00	MXN	56.43						
\$	5.00	MXN	94.05						
\$	6.00	MXN	112.86						
\$	7.00	MXN	131.67						
\$	8.00	MXN	150.48						
\$	9.00	MXN	169.29						
\$	11.00	MXN	206.91						

Step 1: Sample or population?

It is a sample => we have to use the sample formulas

Step 2: Find the mean

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Standard deviation and coefficient of variation

Pizza price example

	NY Dollars		Pesos		Dollars		Pesos		
	\$	1.00	MXN	18.81	Mean	\$	5.50	MXN	103.48
	\$	2.00	MXN	37.62	Sample variance	\$²	10.72	MXN²	3793.69
	\$	3.00	MXN	56.43					
	\$	3.00	MXN	56.43					
	\$	5.00	MXN	94.05					
	\$	6.00	MXN	112.86					
	\$	7.00	MXN	131.67					
	\$	8.00	MXN	150.48					
	\$	9.00	MXN	169.29					
	\$	11.00	MXN	206.91					

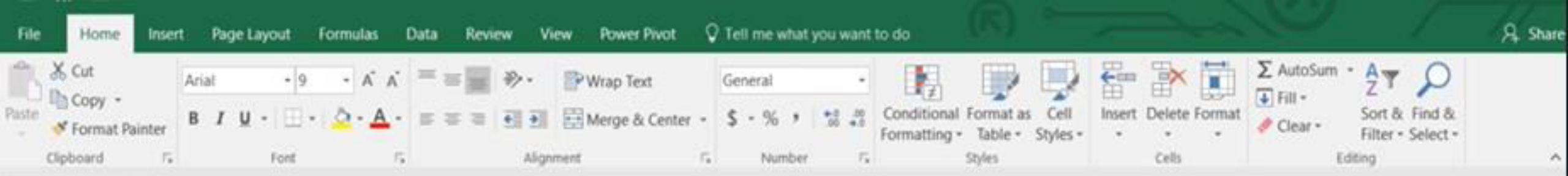
Sample variance

$$\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

Step 1: Sample or population?

Step 2: Find the mean

Step 3: Find the sample variance



A1 X Y Z

1 Standard deviation and coefficient of variation
2 Pizza price example

NY Dollars		Pesos	
\$	1.00	MXN	18.81
\$	2.00	MXN	37.62
\$	3.00	MXN	56.43
\$	3.00	MXN	56.43
\$	5.00	MXN	94.05
\$	6.00	MXN	112.86
\$	7.00	MXN	131.67
\$	8.00	MXN	150.48
\$	9.00	MXN	169.29
\$	11.00	MXN	206.91

	Dollars		Pesos	
Mean	\$	5.50	MXN	103.46
Sample variance	\$ ²	10.72	MXN ²	3793.69
Sample standard deviation	\$	3.27	MXN	61.59

Sample standard deviation

$$\sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

Step 1: Sample or population?

Step 2: Find the mean

Step 3: Find the sample variance

Step 4: Find the sample standard deviation

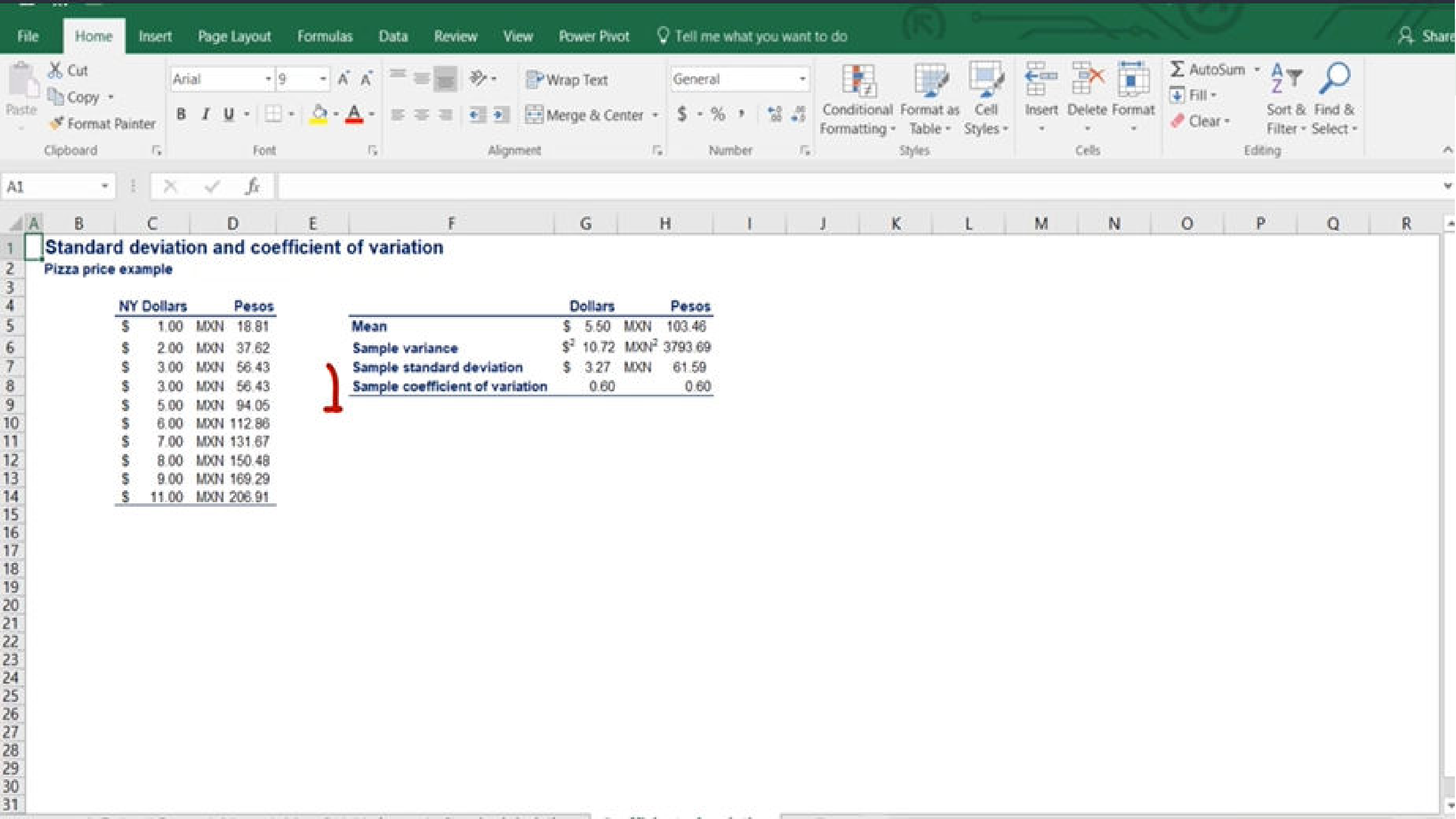


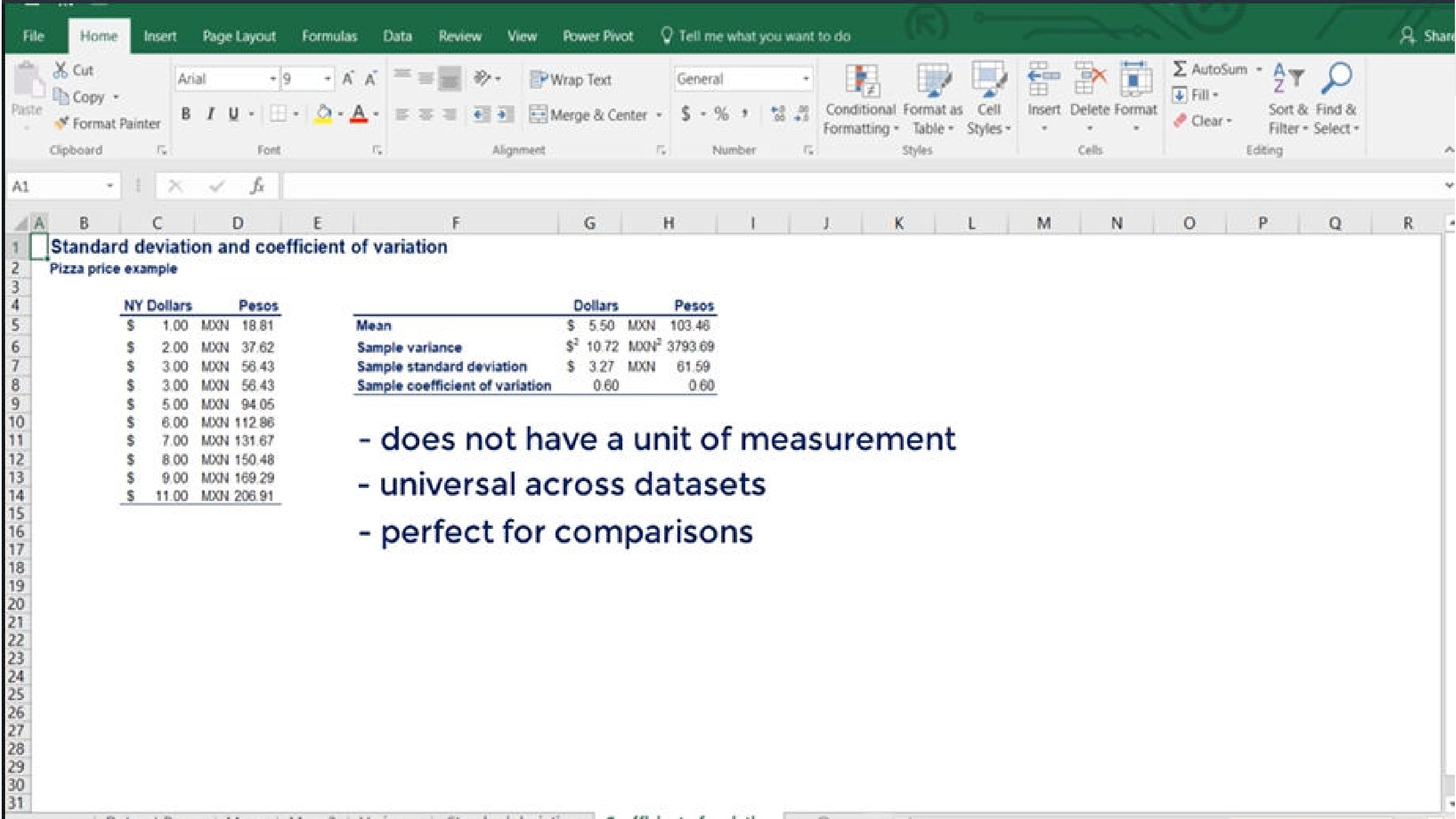
Pizza price example

NY Dollars	Pesos
\$ 1.00	MXN 18.81
\$ 2.00	MXN 37.62
\$ 3.00	MXN 56.43
\$ 3.00	MXN 56.43
\$ 5.00	MXN 94.05
\$ 6.00	MXN 112.86
\$ 7.00	MXN 131.67
\$ 8.00	MXN 150.48
\$ 9.00	MXN 169.29
\$ 11.00	MXN 206.91

	Dollars	Pesos
Mean	\$ 5.50	MXN 103.46
Sample variance	\$ ² 10.72	MXN ² 3793.69
Sample standard deviation	\$ 3.27	MXN 61.59

Standard deviation is the preferred measure of variability, as it is directly interpretable





σ^2

variance

σ

standard deviation

c_v

coefficient of
variation

RECAP



Question 1:

Johnny wanted to know how many days per week people in his class exercise, so he asked 3 of his friends. The answers he got were 1, 3 and 5. Based on this, which of the following is true about the results Johnny found?

Hint: Do the numbers [1,3,5] represent a population or a sample?

☐ $\mu = 3, \sigma = 2$

☐ $\bar{x} = 3, \sigma = -2$

☐ $\bar{x} = 3, s = 2$

☐ $\bar{x} = 3, s = 4$



An illustration of a notepad with a fountain pen. The notepad is tilted and has a checklist of univariate measures. The background is a teal surface with a white geometric shape in the top left and a purple geometric shape in the top right. The notepad is yellow with black dashed lines. A blue fountain pen with a yellow band is lying on the notepad. The checklist items are: Univariate measures, Central tendency, Asymmetry, and Variability, each followed by a large orange checkmark.

Univariate measures

Central tendency ✓

Asymmetry ✓

Variability ✓

in the next two
lessons...

MEASURES OF RELATIONSHIP BETWEEN VARIABLES



Univariate measures

Central tendency ✓

Asymmetry ✓

Variability ✓

in the next two
lessons...

MEASURES OF RELATIONSHIP BETWEEN VARIABLES



OUR FOCUS:

Covariance

Linear correlation
coefficient

REAL ESTATE

what determines house prices?



REAL ESTATE

Their size!



22. Calculating and understanding covariance

Press Esc to exit full screen

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Wrap Text

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Conditional Formatting

Format as Table

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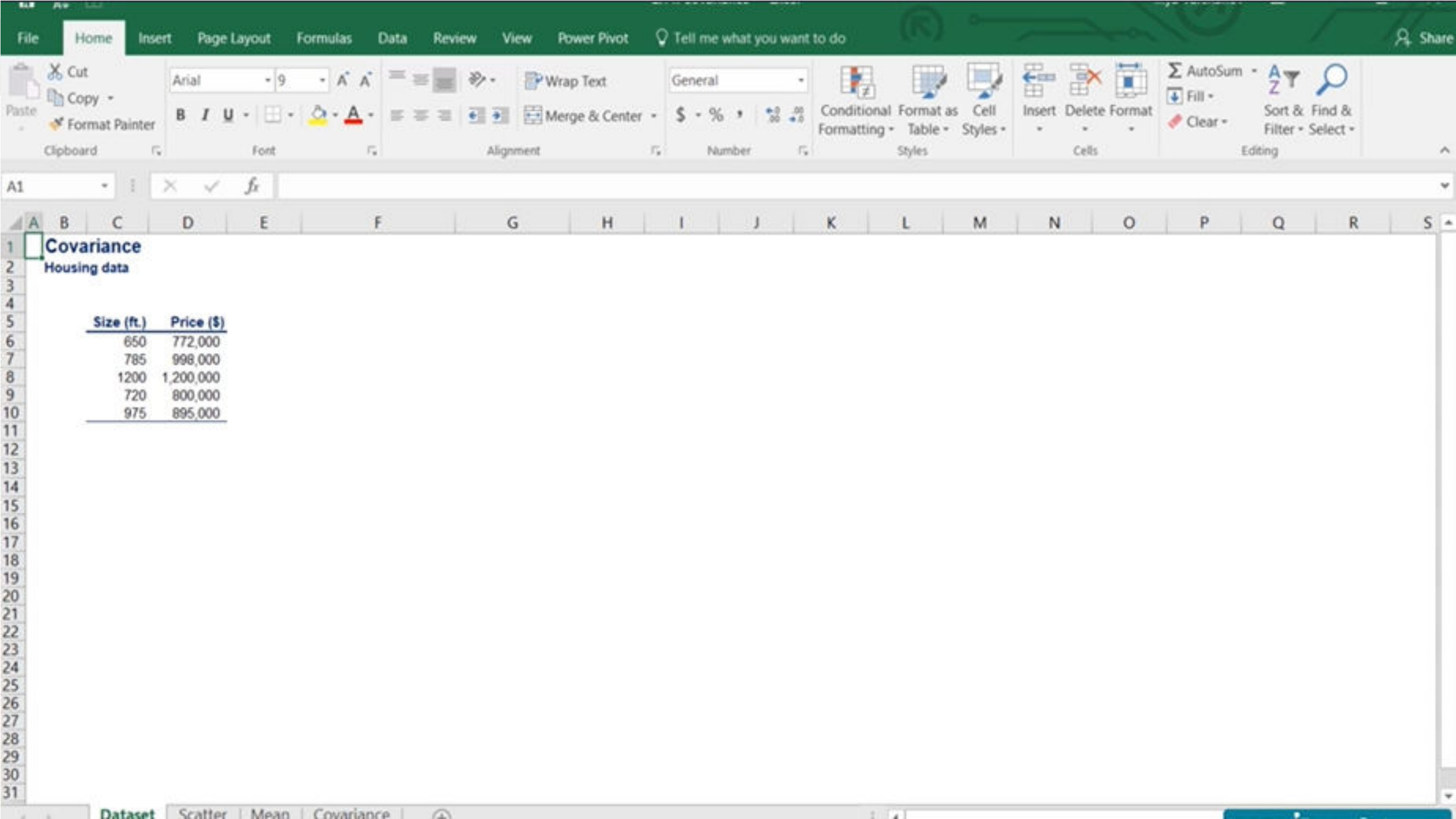
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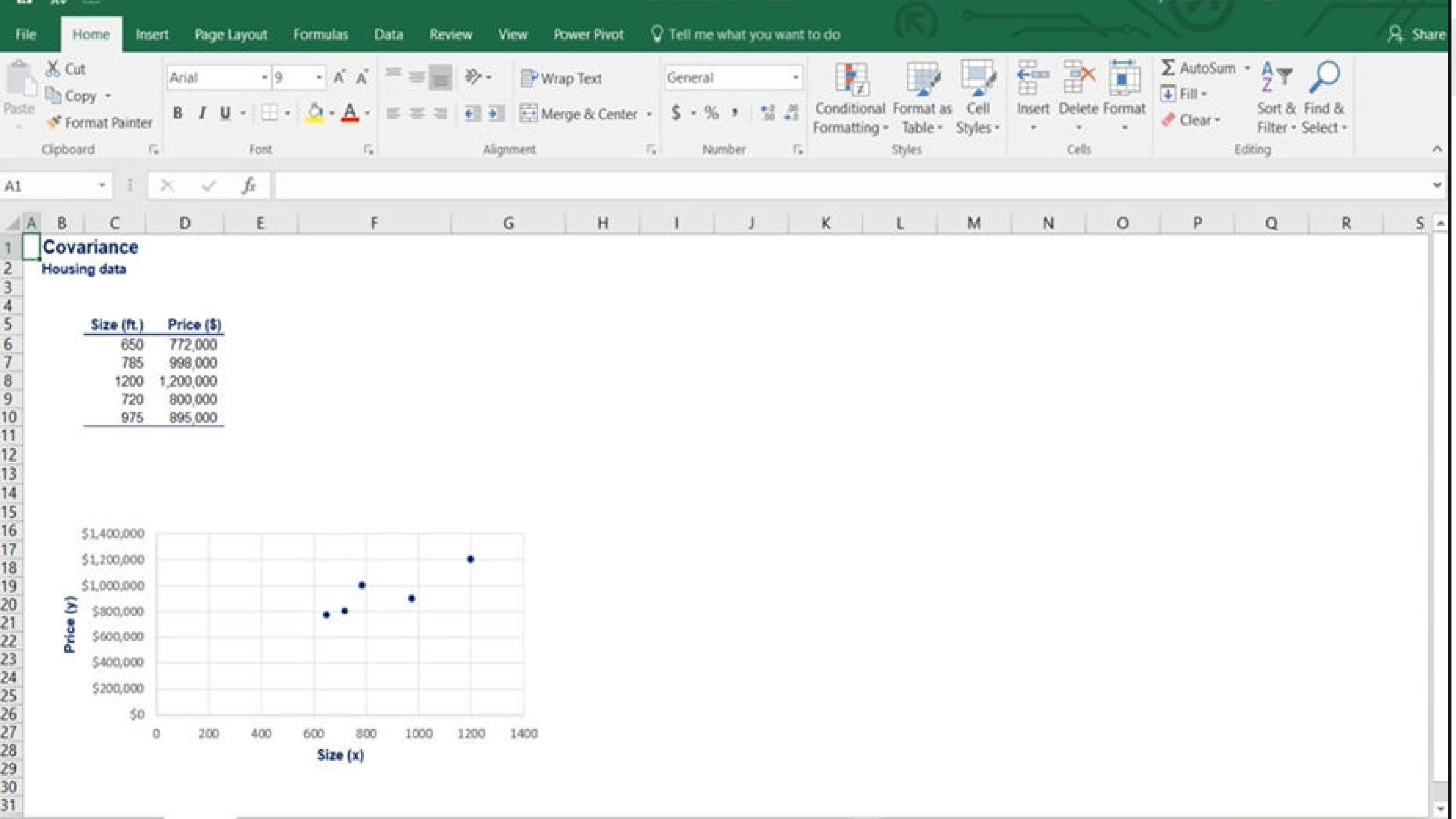
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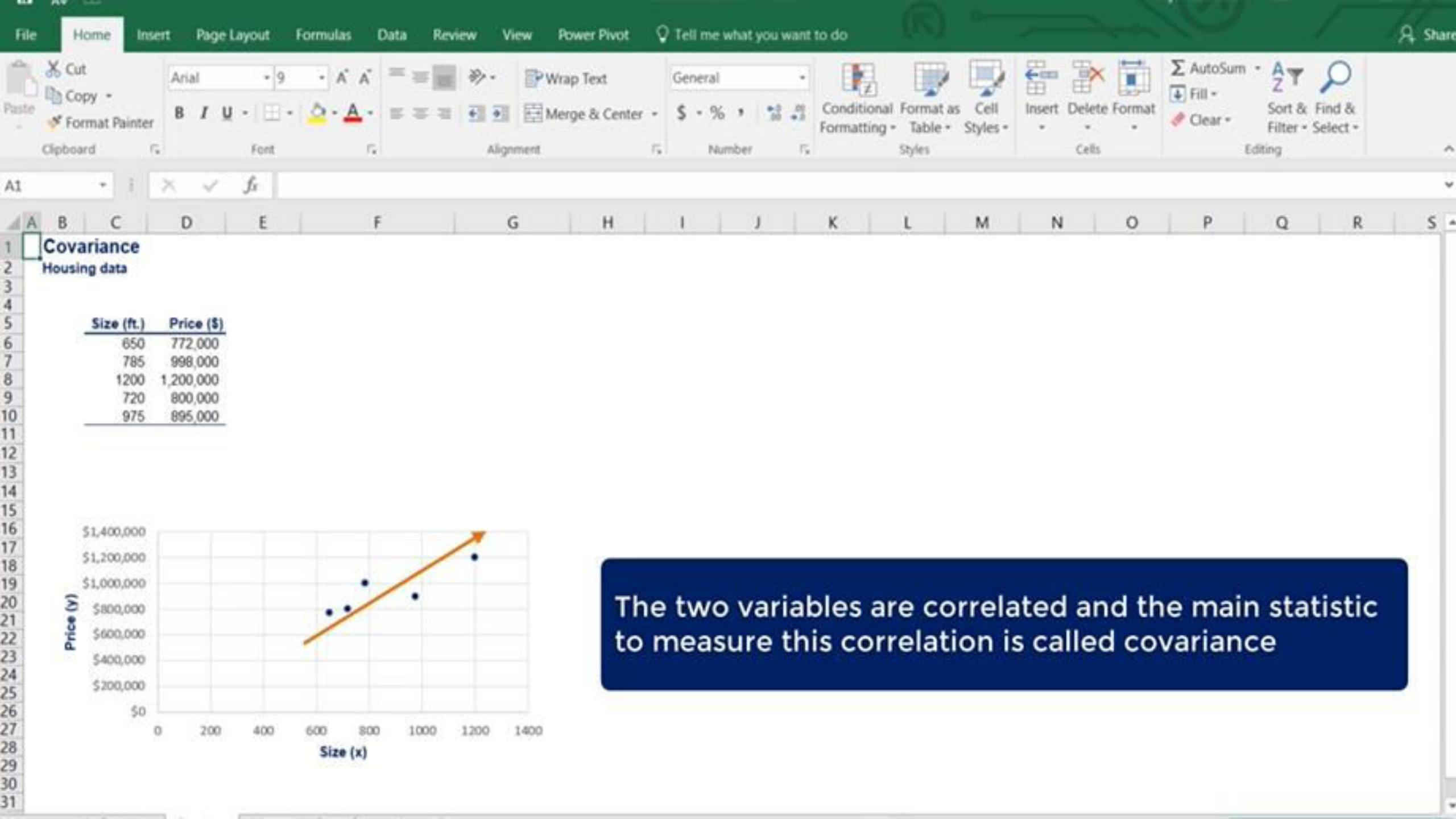
Covariance

Housing data

Size (ft.)	Price (\$)
650	772,000
785	998,000
1200	1,200,000
720	800,000
975	895,000



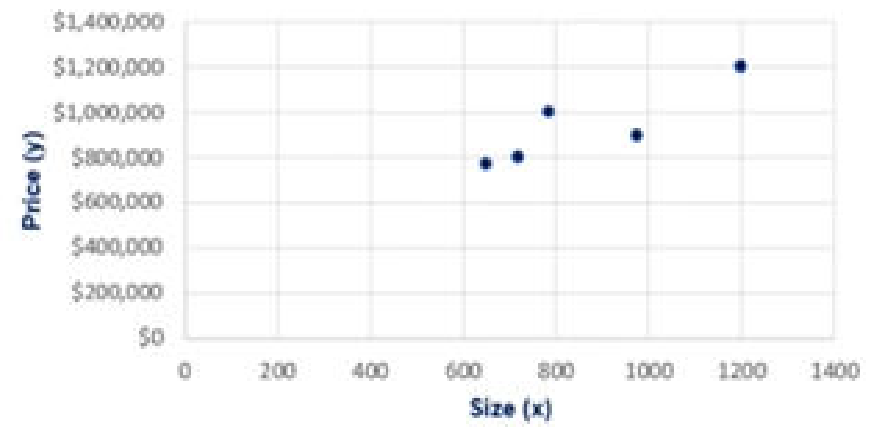




Covariance

Housing data

Size (ft.)	Price (\$)
650	772,000
785	998,000
1200	1,200,000
720	800,000
975	895,000



Covariance may be:

- > 0
- $= 0$
- < 0

1 Covariance
2 Housing data

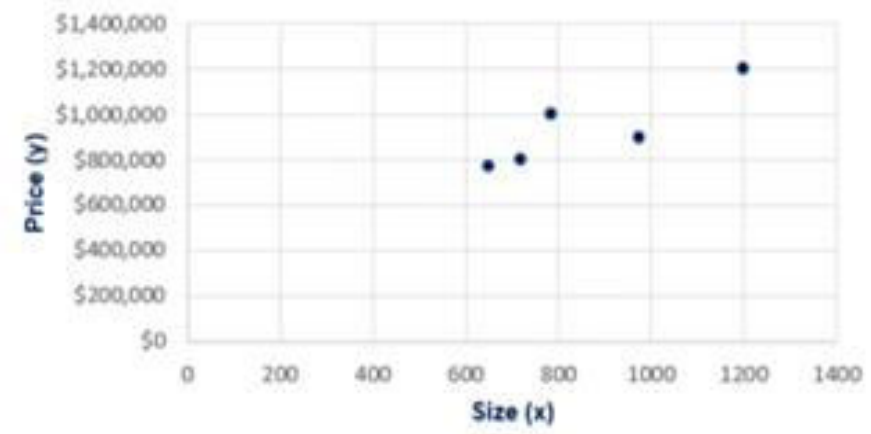
Size (ft.)	Price (\$)
650	772,000
785	998,000
1200	1,200,000
720	800,000
975	895,000

Sample formula

$$s_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x}) * (y_i - \bar{y})}{n-1}$$

Population formula

$$\sigma_{xy} = \frac{\sum_{i=1}^N (x_i - \mu_x) * (y_i - \mu_y)}{N}$$



Covariance

Housing data

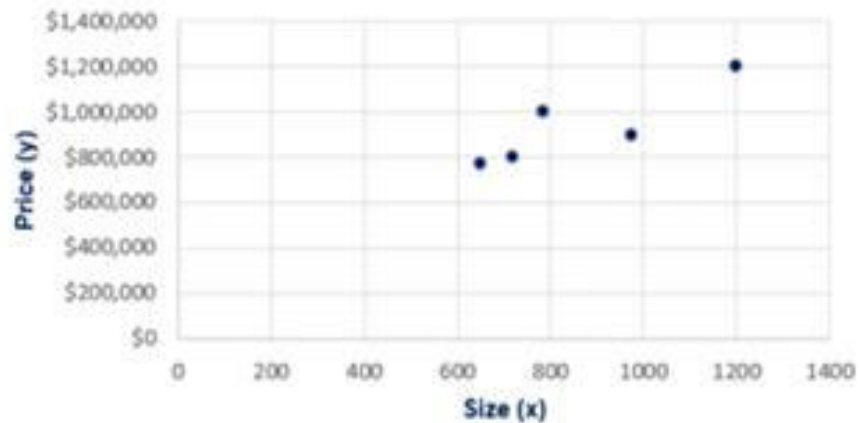
Size (ft.)	Price (\$)
650	772,000
785	998,000
1200	1,200,000
720	800,000
975	895,000

Sample formula

$$s_{xy} = \frac{\sum_{i=1}^n (x_i - \bar{x}) * (y_i - \bar{y})}{n-1}$$

Population formula

~~$$\sigma_{xy} = \frac{\sum_{i=1}^N (x_i - \mu_x) * (y_i - \mu_y)}{N}$$~~



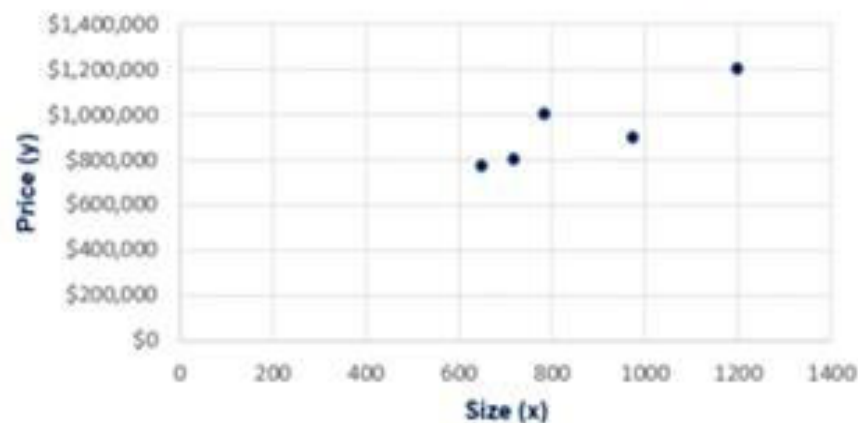
Scatter plot showing Price (y) vs Size (x) for 5 houses. The x-axis ranges from 0 to 1400, and the y-axis ranges from \$0 to \$1,400,000. The data points are approximately:

Size (x)	Price (y)
650	\$750,000
700	\$800,000
750	\$1,000,000
950	\$900,000
1200	\$1,200,000

Covariance

Housing data

	x	y	
	Size (ft.)	Price (\$)	$(x-\bar{x})*(y-\bar{y})$
	650	772,000	34,776,000
	785	998,000	- 5,265,000
	1200	1,200,000	89,178,000
	720	800,000	19,418,000
	975	895,000	- 4,142,000
Mean	866	933,000	133,965,000
Sum			5
Sample size			33,491,250
Cov. Sample			



Covariance

Housing data

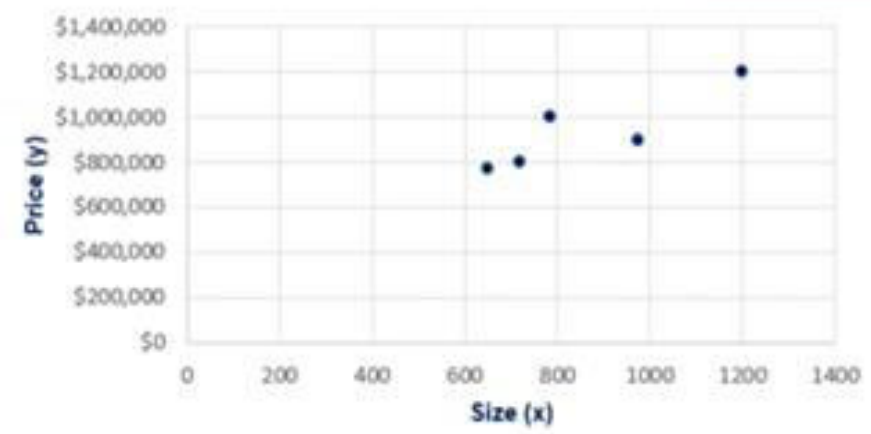
	x	y	
	Size (ft.)	Price (\$)	$(x-\bar{x})*(y-\bar{y})$
	650	772,000	34,776,000
	785	998,000	- 5,265,000
	1200	1,200,000	89,178,000
	720	800,000	19,418,000
	975	895,000	- 4,142,000
Mean	866	933,000	133,965,000
Sum			5
Sample size			33,491,250
Cov. Sample			

Covariance gives a sense of direction

> 0, the two variables move together

< 0, the two variables move in opposite directions

= 0, the two variables are independent



NEXT LESSON: CORRELATION COEFFICIENT

1 Correlation coefficient

2 Housing data

3

4

	Size (ft.)	Price (\$)		$(x-\bar{x})*(y-\bar{y})$
6	650	772,000		34,776,000
7	785	998,000		- 5,265,000
8	1200	1,200,000		89,178,000
9	720	800,000		19,418,000
10	975	895,000		- 4,142,000
11	Mean	866	933,000	Sum 133,965,000
12	Standard dev.	222	173,615	Sample size 5
13				Cov. Sample 33,491,250

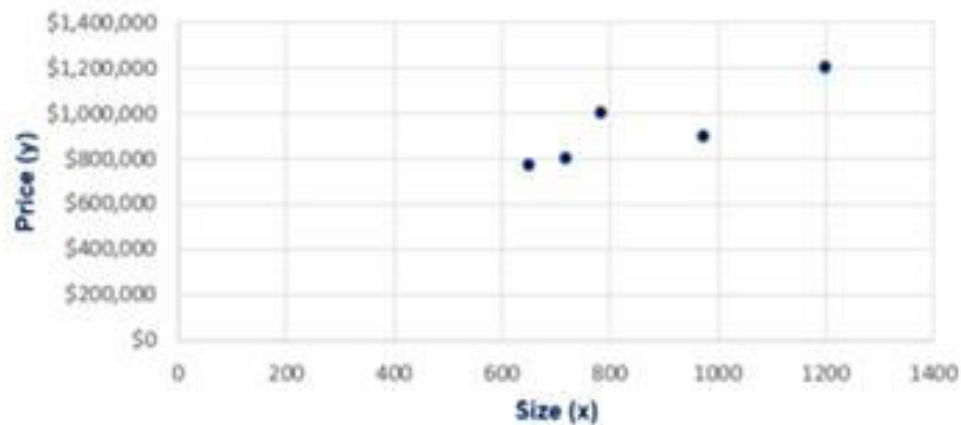
$$\frac{Cov(x, y)}{Stdev(x) * Stdev(y)}$$



Correlation coefficient

Housing data

Size (ft.)	Price (\$)	
650	772,000	
785	998,000	
1200	1,200,000	
720	800,000	
975	895,000	
Mean	900	923,000
Standard dev.	222	173,615
Sum		133,965,000
Sample size		5
Cov. Sample		33,491,250



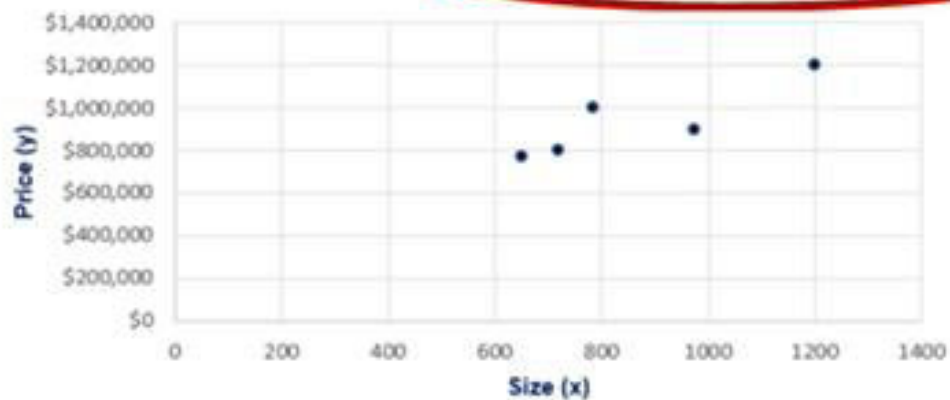
$$\frac{Cov(x, y)}{Stdev(x) * Stdev(y)}$$

$$\frac{S_{xy}}{S_x S_y} \quad \frac{\sigma_{xy}}{\sigma_x \sigma_y}$$

Correlation coefficient

Housing data

	Size (ft.)	Price (\$)		$(x-\bar{x})*(y-\bar{y})$
	650	772,000		34,776,000
	785	998,000		-5,265,000
	1200	1,200,000		89,178,000
	720	800,000		19,418,000
	975	895,000		-4,142,000
Mean	866	933,000	Sum	133,965,000
Standard dev.	222	173,615	Sample size	5
			Dev. Sample	33,431,850
			Correlation coeff.	0.87

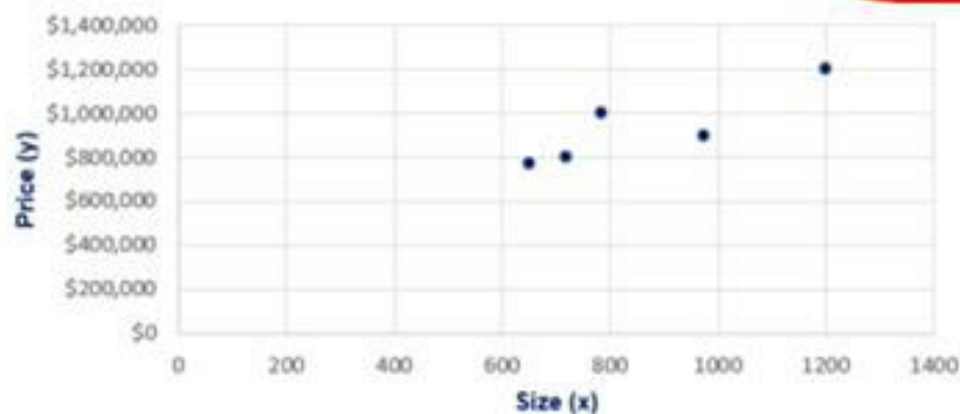


$$-1 \leq \text{correlation coefficient} \leq 1$$

Correlation coefficient

Housing data

	Size (ft.)	Price (\$)		$(x-\bar{x})*(y-\bar{y})$
	650	772,000		34,776,000
	785	998,000		- 5,265,000
	1200	1,200,000		89,178,000
	720	800,000		19,418,000
	975	895,000		- 4,142,000
Mean	866	933,000	Sum	133,965,000
Standard dev.	222	173,615	Sample size	5
			Cov. Sample	33,491,250
			Correlation coeff.	0.87



There is a **STRONG** relationship between the two variables

PERFECT POSITIVE CORRELATION

X **Y**



Correlation coeff. = 1

the entire variability of
one variable is explained
by the other

RELATIONSHIP DIRECTION

Size determines price



CORRELATION OF 0

Absolutely independent variables

CORRELATION OF 0

They have nothing in common!



Coffee in Brazil



Houses in London

NEGATIVE CORRELATION

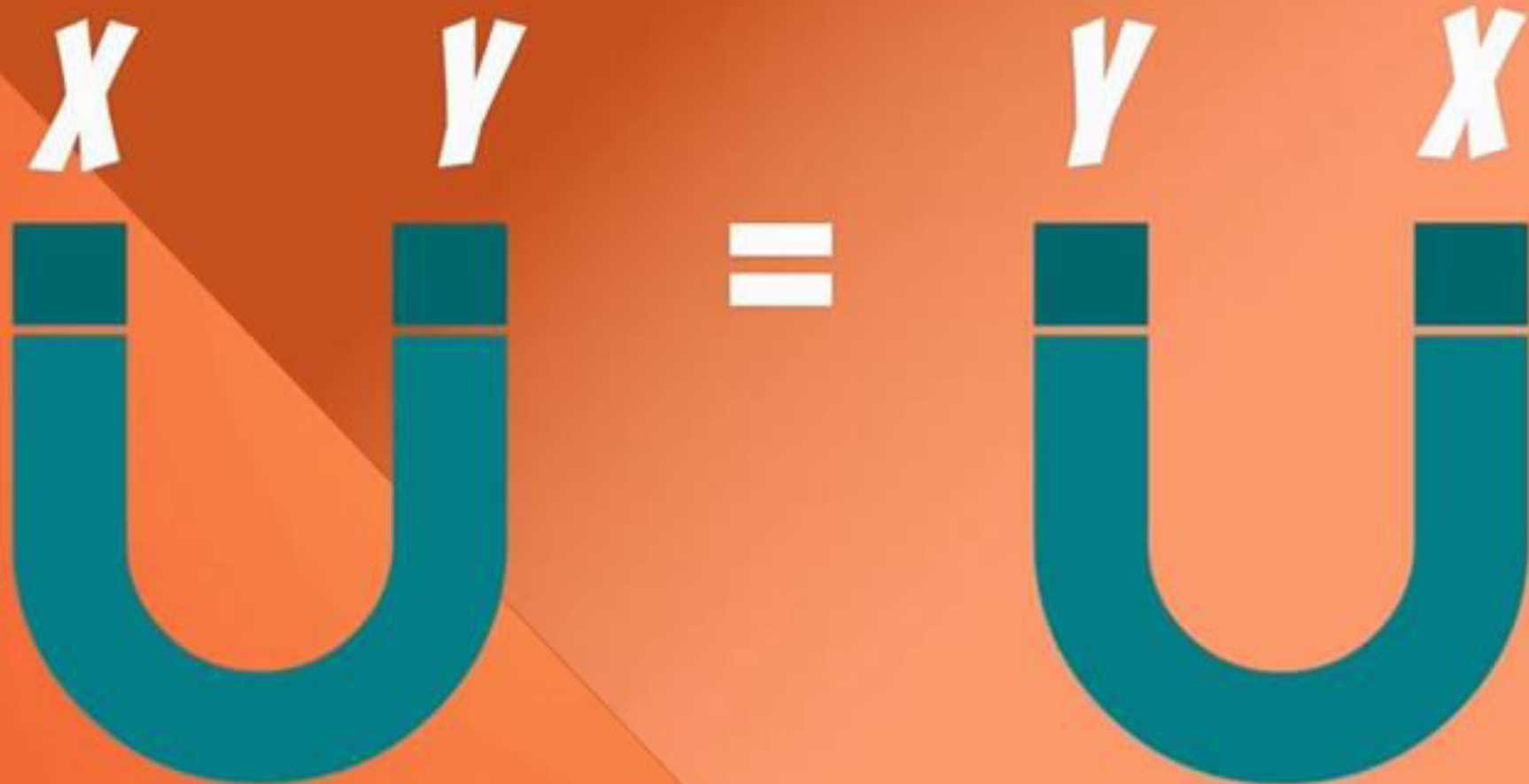
Perfect negative correlation of -1

Imperfect negative correlation: $(-1, 0)$

NEGATIVE CORRELATION



CORRELATION



CORRELATION

$$\frac{\text{Cov}(x, y)}{\text{Stdev}(x) * \text{Stdev}(y)} = \frac{\text{Cov}(y, x)}{\text{Stdev}(y) * \text{Stdev}(x)}$$

Symmetrical with respect to both variables

CAUSALITY

Important to understand the direction of
causal relationships

CAUSALITY

Important to understand the direction of
causal relationships

CORRELATION DOES NOT IMPLY CAUSATION

Question 1:

Which of these is true about correlation and causality?

☐ Causality is a symmetric relation

☐ Correlation is an asymmetric relation (x affects y is different from y affects x)

☐ Causality is an asymmetric relation. (x causes y is different from y causes x)

☐ None of the above

NEW SECTION

PROBABILITY AND INFERENCE

From chance to evidence - one simple step at a time

CHANCE

?

How likely is
an event?

SAMPLES

n

What can a part
tell us?

DECISIONS

✓

Is the evidence
strong enough?

PROBABILITY

WHAT IS PROBABILITY?

Probability measures how likely an event is to happen.



PROBABILITY

THE BASIC PROBABILITY RULE

THE CORE IDEA

$$P(\text{Event}) = \text{Favorable outcomes} / \text{Total possible outcomes}$$

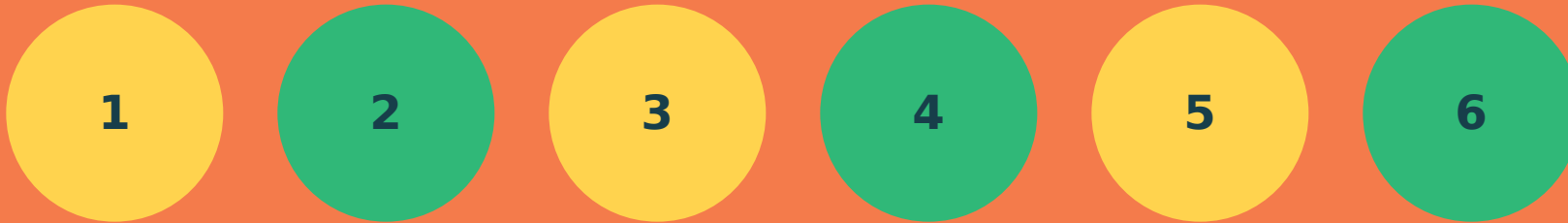
Probability always lies between 0 and 1.

Multiply by 100 to express it as a percentage.

PROBABILITY

EXAMPLE: ROLLING A DIE

What is the probability of rolling an even number?



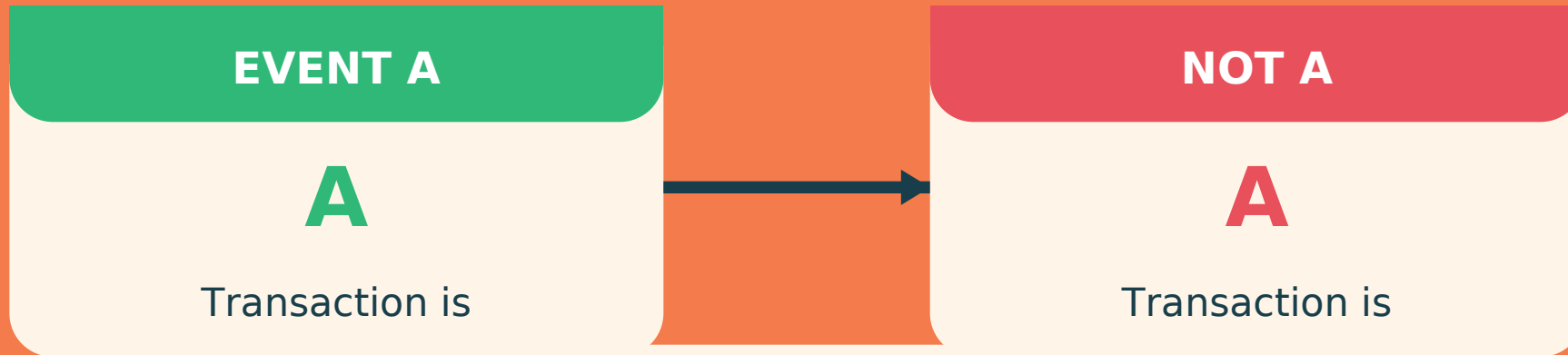
FAVORABLE: 2, 4, 6

$$P(\text{Even}) = 3 / 6 = 1 / 2 = 50\%$$

PROBABILITY

COMPLEMENT RULE

An event either happens or it does not happen.



$$P(\text{not } A) = 1 - P(A)$$

PROBABILITY

JOINT EVENTS: AND / OR

A AND B

AND

Both events happen
 $P(A \text{ and } B)$

A OR B

OR

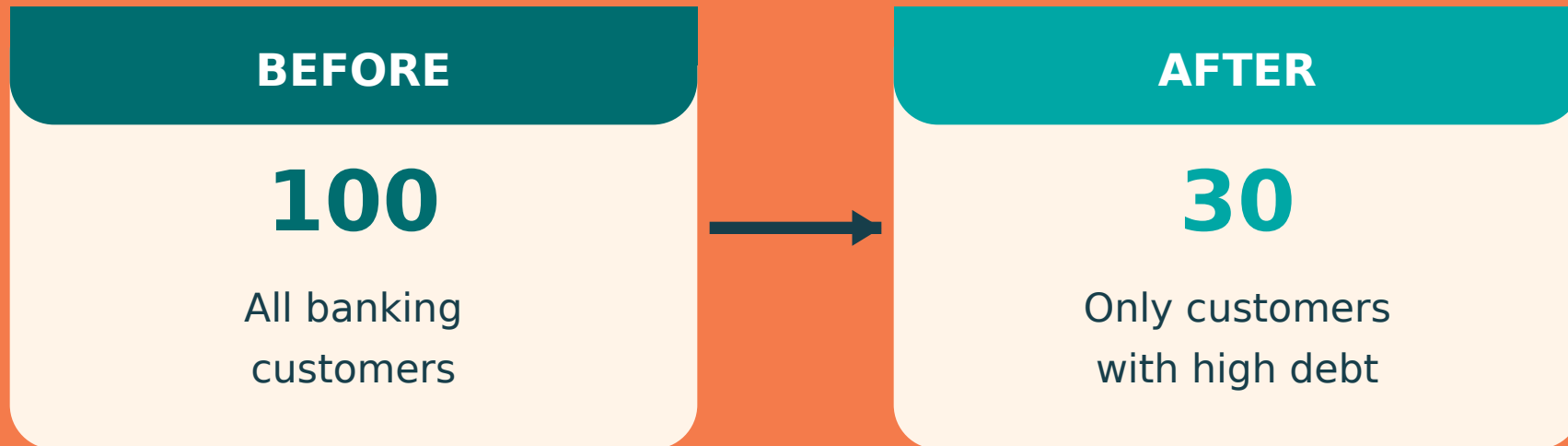
At least one happens
 $P(A \text{ or } B)$

AND narrows the condition. OR widens the condition.

CONDITIONAL PROBABILITY

CONDITIONAL PROBABILITY

Probability after receiving new information



CONDITIONAL PROBABILITY

WHAT DOES $P(A \mid B)$ MEAN?

READ AS: PROBABILITY OF A, GIVEN B

$$P(A \mid B) = P(A \text{ and } B) / P(B)$$

The vertical bar $|$ means "given that".

We restrict our attention to cases where B has happened.

CONDITIONAL PROBABILITY

BANKING EXAMPLE

Among 100 customers, 30 have high debt.
Of those 30 customers, 12 miss a payment.

HIGH DEBT

30

30 customers

MISSED PAYMENT

12

12 customers

$$P(\text{Missed} \mid \text{High debt}) = 12 / 30 = 40\%$$

CONDITIONAL PROBABILITY

INDEPENDENT EVENTS

Independent means one event does not change the other.

EVENT A

A

Coin lands
Heads

EVENT B

B

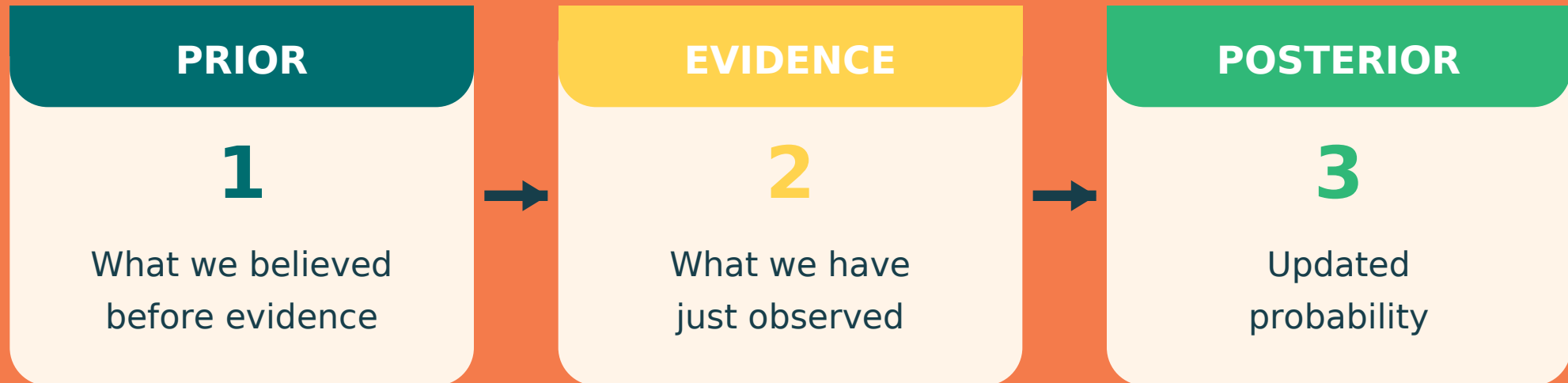
Die shows
6

If independent: $P(A \text{ and } B) = P(A) \times P(B)$

BAYES THEOREM

BAYES' THEOREM

Update a belief when new evidence appears



BAYES THEOREM

BAYES' FORMULA

UPDATE RULE

$$P(A \mid B) = P(B \mid A) \times P(A) / P(B)$$

Prior x likelihood, adjusted by total evidence.

BAYES THEOREM

FRAUD ALERT EXAMPLE

An alert does not automatically mean a transaction is fraud.

BASE RATE

1%

1% of transactions
are fraudulent

DETECTION

90%

90% of fraud
triggers an alert

FALSE ALERT

5%

5% of genuine
transactions alert

BAYES THEOREM

BAYES WITH 10,000 TRANSACTIONS

FRAUD

90

100 fraud
90 alerts

GENUINE

495

9,900 genuine
495 alerts

ALL ALERTS

585

90 + 495
= 585

$$P(\text{Fraud} \mid \text{Alert}) = 90 / 585 = 15.4\%$$

BAYES THEOREM

THE BAYES LESSON

A test can be accurate and still produce many false alarms.

DO NOT ASK

?

"Is the alert
accurate?"

ASK INSTEAD

✓

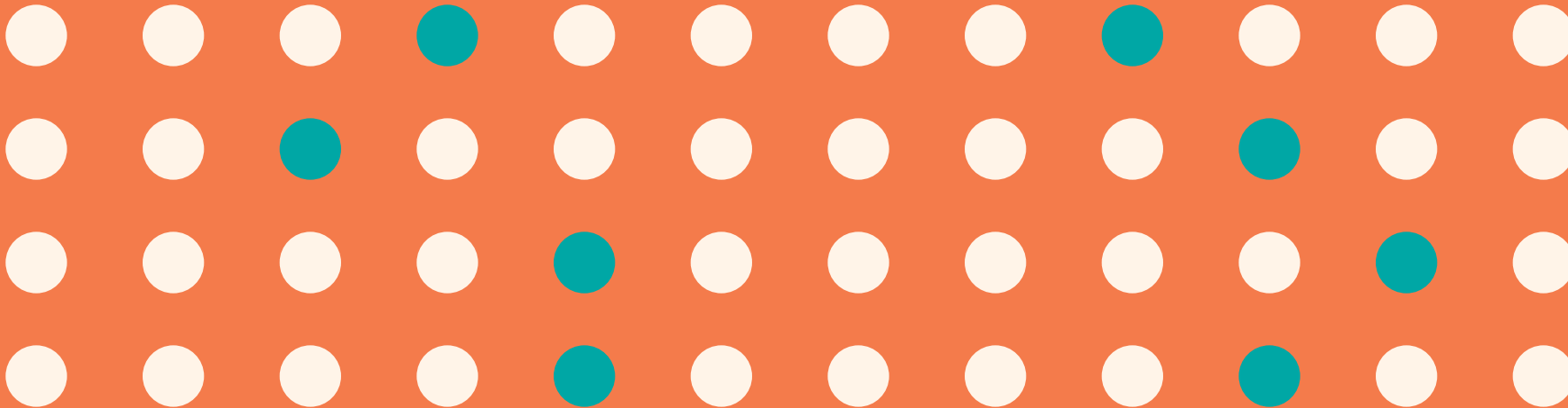
"Given the alert,
what is likely?"

Always consider the base rate.

SAMPLING

SAMPLING

Learn about a large population using a smaller group



Population

 Selected sample

SAMPLING

WHY SAMPLE?

FASTER



Less time to
collect data

CHEAPER



Lower cost than
a full census

PRACTICAL



Often the only
realistic option

SAMPLING

GOOD AND BAD SAMPLES

REPRESENTATIVE



Different regions
Different ages
Random selection

BIASED



Only one branch
Only one age group
Convenience selection

A large biased sample is still biased.

SAMPLING

SAMPLING DISTRIBUTION

Take many samples and calculate the mean of each sample.

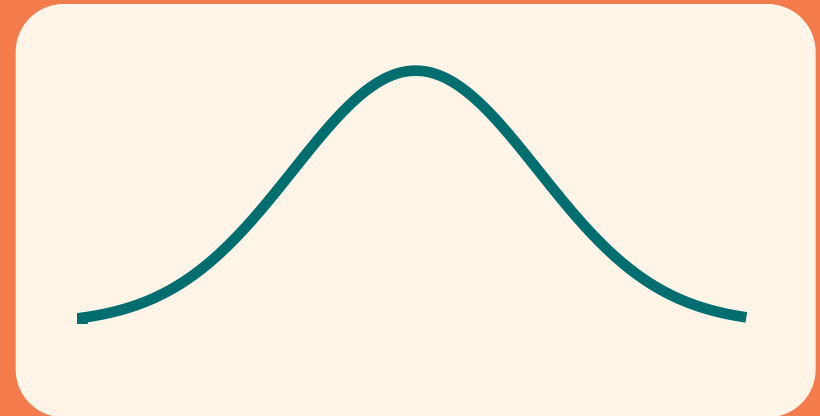
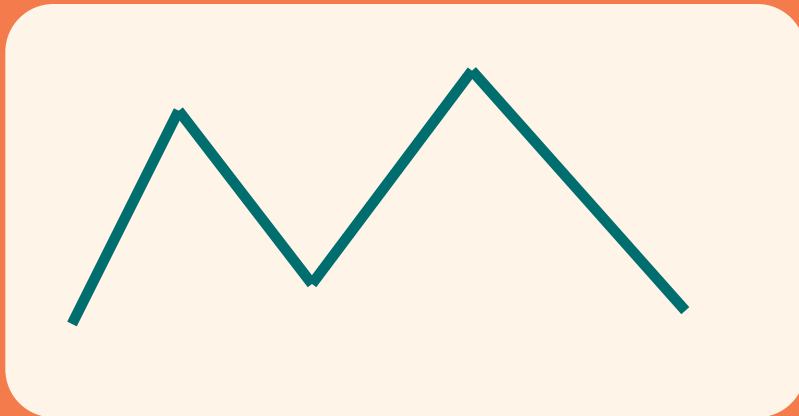


These sample means form a new distribution.

CENTRAL LIMIT THEOREM

CENTRAL LIMIT THEOREM

With sufficiently large random samples...



The distribution of sample means becomes approximately normal.

CENTRAL LIMIT THEOREM

WHAT CLT GIVES US

CENTRE

μ

Mean of sample means
 $\approx$ population mean

SHAPE

n

Approximately normal
for large samples

POWER

✓

Enables confidence
intervals and tests

STANDARD ERROR

STANDARD ERROR

Standard error measures how much sample means vary.

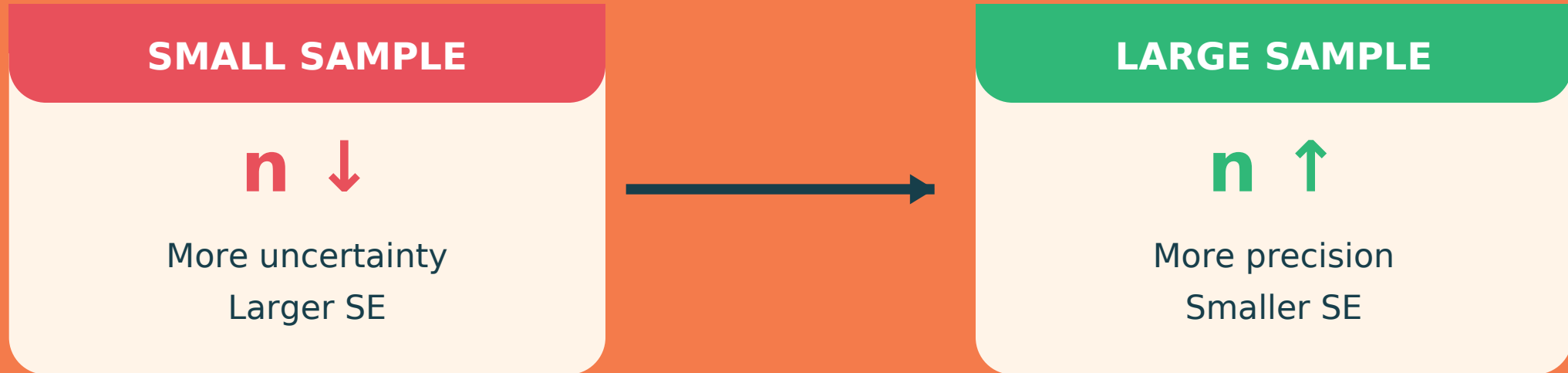
STANDARD ERROR OF THE SAMPLE MEAN

$$SE = \sigma / \sqrt{n}$$

σ = population standard deviation n = sample size

STANDARD ERROR

SAMPLE SIZE AND STANDARD ERROR



When sample size increases, standard error decreases.

HYPOTHESIS TESTING

HYPOTHESIS TESTING

Use sample evidence to evaluate a claim

NULL HYPOTHESIS H_0

H_0

The default claim

No effect

No difference

ALTERNATIVE H_1

H_1

What we look for

An effect

A difference

HYPOTHESIS TESTING

A SIMPLE BANKING CLAIM

Claim: a new process reduces average approval time.

H0

=

Average time has
not decreased

H1

<

Average time
has decreased

Collect a sample, calculate a statistic, then assess the evidence.

HYPOTHESIS TESTING

THE TESTING PROCESS

1

STATE

H0 and H1

2

SAMPLE

Collect data

3

CALCULATE

Test statistic

4

DECIDE

Use p-value

P-VALUES

WHAT IS A P-VALUE?

Assume H_0 is true.

The p-value asks:

**How surprising is our sample result
under that assumption?**

Smaller p-value = stronger evidence against H_0 .

P-VALUES

P-VALUE DECISION RULE

$$p \leq \alpha$$



Reject H_0
Evidence is
statistically significant

$$p > \alpha$$



Do not reject H_0
Evidence is
not strong enough

A common significance level is $\alpha = 0.05$.

P-VALUES

WHAT A P-VALUE DOES NOT MEAN

NOT THIS



Probability that
H0 is true

NOT THIS



Size or importance
of the effect

NOT THIS



Proof that a result
will repeat

A p-value measures compatibility with H0 - not practical importance.

CONFIDENCE INTERVALS

CONFIDENCE INTERVAL

A confidence interval gives a plausible range for a population value.



Estimate $\pm$ Margin of error

CONFIDENCE INTERVALS

95% CONFIDENCE - CORRECT MEANING

**If we repeated the sampling process many times,
about 95% of intervals built this way would
contain the true population value.**

It describes the method, not a probability assigned after calculation.

CONFIDENCE INTERVALS

WHAT CONTROLS INTERVAL WIDTH?

SAMPLE SIZE

n

Larger n gives a narrower interval

VARIABILITY

σ

More variation gives a wider interval

CONFIDENCE

%

Higher confidence gives a wider interval

INFERENCE

CONFIDENCE INTERVAL AND TEST

For a two-sided test at $\alpha = 0.05$:

NULL VALUE OUTSIDE

OUT

Reject H_0
 $p < 0.05$

NULL VALUE INSIDE

IN

Do not reject H_0
 $p \geq 0.05$

The interval also shows the estimated effect and its precision.

RECAP

THE COMPLETE PICTURE

Describe → sample → infer → decide

1

PROBABILITY

Quantify chance

2

SAMPLING

Observe a part

3

CLT + SE

Measure uncertainty

4

CI + TEST

Evaluate evidence